Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

PHYSICS

Class – 9

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1. MOTION

Solutions

Worksheet – 1

1. **Correct answer: (B) Kinematics**

   Explanation: Kinematics deals with motion without considering forces acting on bodies. (No force term used)

2. **Correct answer: (B) in motion**

   Explanation: Motion is relative; with respect to a stationary observer, the car's position changes with time.

3. **Correct answer: (C) they depend on the observer**

   Explanation: Directions like left and right are relative and depend on the observer's frame of reference.

4. **Correct answer: (C) place**

   Explanation: Day and night depend on the observer's position on Earth due to Earth's rotation.

5. **Correct answer: (B) straight line**

   Explanation: For uniform motion, speed is constant: $v = \frac{d}{t} = \text{constant} \Rightarrow d \propto t$
   Hence, the distance–time graph is a straight line.

6. **Correct answer: (C) 5 m/s**

   Explanation: Speed $= \frac{\text{Distance}}{\text{Time}} = \frac{100}{20} =$ 5 m/s

7. **Correct answer: (C) zero**

   Explanation: Displacement = Final position − Initial position
   After one complete round, both positions are the same.

8. **Correct answer: (A) 0.5 m/s east**

   Explanation: Net displacement $= 100 - 60 = 40$ m east
   Total time $= 50 + 30 = 80$ s
   Average velocity $= \frac{40}{80} = 0.5$ m/s east

9. **Correct answer: (B) 1.0 m/s north**

   Explanation: Net displacement $= 120 - 40 = 80$ m north
   Total time $= 60 + 20 = 80$ s
   Average velocity $= \frac{80}{80} = 1.0$ m/s north

10. **Correct answer: (A) at rest**

    Explanation: Relative to the passengers, the position of the train does not change with time. $v_{\text{relative}} = 0$

11. **Correct answer (C) reference frame**

    Explanation: Rest or motion depends on observer's reference frame; relative velocity changes with frame ($v_{(\text{relative})} = v_1 - v_2$).

12. **Correct answer: (C) $|\vec{A}|$**

    Explanation: Magnitude of a vector is written using modulus notation $|\vec{A}|$; gives scalar length = $\sqrt{(Ax^2 + Ay^2 + Az^2)}$.

13. **Correct answer: (C) Velocity — Vector**

    Explanation: Velocity has magnitude and direction → vector quantity (v = displacement/time).

Worksheet -2

1. **Correct answer: (C) A car moving on a curved path at constant speed**

   Explanation: Velocity changes due to change in direction, so acceleration is non-uniform (vector changes). $\vec{a} = \frac{d\vec{v}}{dt}$

2. **Correct answer: (C) Final velocity**

   Explanation: Equation directly gives final velocity. $v = u + at$

3. **Correct answer: (C) $\boldsymbol{v = u + at}$**

   Explanation: Known values: u, a, t; required quantity is v. $v = u + at$

4. **Correct answer: (B) 2 m s^{-2}**

   Explanation: $a = \frac{v-u}{t} = \frac{30-10}{10} = 2$ m s^{-2}

5. **Correct answer: (C) 2 m s^{-2}**

   Explanation: $a = \frac{25-5}{10} = 2$ m s^{-2}

6. **Correct answer: (B) 50 m**

   Explanation: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 4 \times 25 = 50$ m

7. **Correct answer: (C) 40 m**

   Explanation: $v^2 = u^2 + 2as$

   $0 = 20^2 + 2(-5)s \Rightarrow s = 40$ m

8. **Correct answer: (C) 37.5 m**

   Explanation: $0 = 15^2 + 2(-3)s \Rightarrow s = \frac{225}{6} = 37.5$ m

9. **Correct answer: (C) 20 m s^{-1}**

   Explanation: $s = \frac{1}{2}at^2 \Rightarrow a = \frac{2s}{t^2} = \frac{80}{16} = 5$

   $v = at = 5 \times 4 = 20$ m s^{-1}

10. **Correct answer: (B) 20 m**

    Explanation: Distance in nth second: s_n = u + (a/2)(2n−1). Here u=2, a=4, n=5 → s_5 = 2 + 2(9) = 20 m. Correct answer 20 m

11. **Correct answer: (A) 15**

    Explanation: s_n = u + (a/2)(2n−1). From rest u=0, a=2 → s_n = (2/2)(2n−1)=2n−1 = 29 ⇒ n = 15.

Worksheet -3

1. **Correct answer: (C) Cyclist moving with constant speed on a straight path**

   Explanation: A horizontal line on a velocity–time graph means constant velocity. $v =$ constant

2. **Correct answer: (C) A has uniform velocity, B has increasing velocity**

   Explanation: Straight-line $s-t$ graph → constant velocity; curved upward → velocity increasing with time. $v = \frac{ds}{dt}$

3. **Correct answer: (C)**

4. **Correct answer: (A) Velocity × time equals distance**

   Explanation: Area under $v-t$ graph equals displacement. $s = \int v\,dt$

5. **Correct answer: (C) 60 m**

   Explanation: $s = vt = 6 \times 10 = 60$ m

6. **Correct answer: (A) 100 m**

   Explanation: $s = \frac{(u+v)}{2}t = \frac{(0+20)}{2} \times 10 = 100$ m

7. **Correct answer: (A) −4 m/s^2**

   Explanation: $a = \frac{v-u}{t} = \frac{0-20}{5} = -4$ m/s^2

8. **Correct answer: (B) 12 m/s**

   Explanation: $\Delta v = at = 3 \times 4 = 12$ m/s

9. **Correct answer: (B) Displacement is zero**

   Explanation: Equal forward and backward motion cancels displacement. Net displacement $= 0$

Worksheet-4

1. **Correct answer: (C) An athlete running on a circular track at constant speed**

   Explanation: Speed is constant but velocity changes due to change in direction. $\vec{a} = \frac{d\vec{v}}{dt} \neq 0$

2. **Correct answer: (C) Speed**

   Explanation: In uniform circular motion, magnitude of velocity (speed) remains constant. $v =$ constant

3. **Correct answer: (C) The direction of motion changes continuously**

   Explanation: Acceleration exists due to continuous change in direction of velocity. $\vec{a} = \frac{d\vec{v}}{dt}$

4. **Correct answer: (C) Athlete running on a circular track**

   Explanation: Speed is constant but direction keeps changing.

   $|\vec{v}| =$ constant, $\vec{v}$ changes

5. **Correct answer: (B) Motion on a curved path**

   Explanation: Direction changes even if speed is constant, so acceleration exists. $a = \frac{v^2}{r}$

6. **Correct answer: (D) Circular track**

   Explanation: Direction of velocity changes continuously (maximum times) on a circular path. Direction change → ∞

7. **Correct answer: (C) π m s^{-1}**

   Explanation: $v = \frac{2\pi r}{T} = \frac{2\pi \times 7}{14} = \pi$ m/s

8. **Correct answer: (A) 2πm s^{-1}**

Explanation: $v = \frac{2\pi r}{T} = \frac{2\pi \times 21}{21} = 2\pi$ m/s

9. **Correct answer: (C) Equal to diameter of the circle**

Explanation: After half revolution, displacement equals straight-line distance between opposite points. Displacement $= 2r$

10. **Correct answer: (B) $\sqrt{2}\,r$**

Explanation: Displacement is chord for 90° angle. $d = \sqrt{r^2 + r^2} = \sqrt{2}\,r$

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (A) −3 m/s²**

Explanation: $a = \frac{v-u}{t} = \frac{10-25}{5} = -3$ m/s^2

2. **Correct answer: (B) 40 km/hr**

Explanation: Speed $= \frac{\text{Distance}}{\text{Time}} = \frac{200}{5} =$ 40 km/hr

3. **Correct answer: C) $\frac{5v_1v_2}{2v_2+3v_1}$**

Explanation:

Average Speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

Let the total distance be D.

Distance covered with speed $v_1 = \frac{2}{5}D$

Distance covered with speed $v_2 = \frac{3}{5}D$

Time taken,

$$t_1 = \frac{\frac{2}{5}D}{v_1} = \frac{2D}{5v_1} \quad t_2 = \frac{\frac{3}{5}D}{v_2} = \frac{3D}{5v_2}$$

Total time $= \frac{2D}{5v_1} + \frac{3D}{5v_2}$; $v_{\text{avg}} = \frac{D}{\frac{2D}{5v_1}+\frac{3D}{5v_2}} = \frac{1}{\frac{2v_2+3v_1}{5v_1v_2}} = \frac{5v_1v_2}{2v_2+3v_1}$

4. **Correct answer: (B) 40 km/hr towards East**

Explanation: Velocity $= \frac{200}{5} =$ 40 km/hr (direction East)

5. **Correct answer: 30 m/s**

Explanation: Speed $= \frac{150}{5} = 30$ m/s

6. **Correct answer: (A) 4 sec**

Explanation: $v = u + at \Rightarrow 0 = 20 - 5t \Rightarrow t = 4$ s

7. **Correct answer: D) 2 m/s²**

Explanation: $a = \frac{v-u}{t} = \frac{6-0}{3} = 2$ m/s^2

8. **Correct answer: A) 0 m/s²**

Explanation: Constant velocity $\Rightarrow a =$ 0(no change in velocity)

9. **Correct answer: D) 18 m/sec**

Explanation: $v = u + at = 10 + (2 \times 4) = 18$ m/s

10. **Correct answer: A) $\frac{7}{5}$**

Explanation: Distance in n^{th}second: $s_n = \frac{a}{2}(2n-1)$.

Ratio $s_4 : s_3 = 7 : 5$

11. **Correct answer: B) 54**

Explanation: km/h = m/s $\times \frac{18}{5} = 15 \times \frac{18}{5} = 54$

12. **Correct answer: C) 24 m**

Explanation: Displacement = area under displacement–time graph. From graph, final displacement at $t = 6\,s$is **24 m**.

13. **Correct answer: A) 45 m**

Explanation: Distance in last second:

$$s_n = \frac{g}{2}(2t-1)$$

$$25 = 5(2t-1) \Rightarrow t = 3\,s$$

Total height: $h = \frac{1}{2}gt^2 = 5 \times 9 = 45\,m$

14. **Correct answer: B) 15 m/s**

Explanation:

First stone: $20 = \frac{1}{2}gt^2 \Rightarrow t = 2\,s$

Second stone falls for $1\,s$: $20 = ut + \frac{1}{2}gt^2$

$$20 = u(1) + 5 \Rightarrow u = 15\,m/s$$

15. **Correct answer: C) 3.5 s**

Explanation: Distance in n^{th}second: $s_n = \frac{g}{2}(2n-1)$.

$$\frac{s_t}{s_{t-1}} = \frac{2t-1}{2t-3} = \frac{3}{2} \Rightarrow 4t - 2 = 6t - 9 \Rightarrow t = 3.5\,s$$

16. **Correct answer: (A) 58.8**

17. **Correct answer: D) 2 m/s²**

Explanation: $a = \frac{v-u}{t} = \frac{6-0}{3} = 2$ m/s^2

18. **Correct answer:**
Explanation: $a = \frac{v-u}{t} = \frac{-2-10}{4} = -3\ \text{m/s}^2$

19. **Correct answer: A) 4 m/s**
Explanation:
Let total distance $= 3d$.
First $d$at 4 m/s: $t_1 = \frac{d}{4}$
Remaining $2d$covered in equal times with 2 m/sand 6 m/s: $2t + 6t = 8t = 2d \Rightarrow t = \frac{d}{4}$
Total time: $t = \frac{d}{4} + \frac{d}{4} + \frac{d}{4} = \frac{3d}{4}$
Average velocity: $v_{avg} = \frac{3d}{3d/4} = 4$ m/s

20. **Correct answer: B) 2.0 m/s**
Explanation: Average velocity $= \frac{\text{displacement}}{\text{time}}$.
Displacement from A to B = diameter $= 2r = 2$ m. ; $v_{avg} = \frac{2}{1} = 2$ m/s

21. **Correct answer:**
A) Its maximum speed is 8 m s^{-1}
C) It travelled a total distance of 24 m
Explanation: Let acceleration time be $t_1$and retardation time be t_2.
Given, $t_1 + t_2 = 6$
Maximum speed, $v_{\max} = a_1 t_1 = a_2 t_2$
$2t_1 = 4t_2 \Rightarrow t_1 = 2t_2$
Substituting, $2t_2 + t_2 = 6 \Rightarrow 3t_2 = 6 \Rightarrow t_2 = 2$ s
$t_1 = 4$ s
$v_{\max} = 2 \times 4 = 8\ \text{m s}^{-1}$
(Option A)
Total distance travelled =
Area of v–t graph $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 6 \times 8 = 24$ m
(Option C)

22. **Correct answer: A) Its displacement is directly proportional to the square of the time.**
D) It always moves in a straight line.
Explanation: For Option A:
Using $s = ut + \frac{1}{2}at^2$and $u = 0, s = \frac{1}{2}at^2 \Rightarrow s \propto t^2$
For Option D:
Since the body starts from rest and moves with uniform (constant) acceleration, the velocity and displacement remain in the direction of acceleration. Hence, the motion is along a straight line.

23. **Correct answer: B) Acceleration is uniform.**
Explanation: The slope of a velocity–time graph represents acceleration. A straight line inclined to the time axis has a constant slope. Hence, acceleration is constant (uniform). Since velocity changes with time, it is not uniform.

24. **Correct answer: D) Started with some initial velocity**
Explanation: Velocity–time graph is a straight line with positive intercept on v-axis ⇒
$u \neq 0, a =$ constant
Velocity is not uniform and acceleration is not zero.

25. **Correct answer: B) 10 m/s²**
Explanation: Uniform acceleration: $a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{45-15}{5-2} = \frac{30}{3} = 10\ \text{m/s}^2$

26. **Correct answer: A**
Explanation: Relative acceleration:
$a_{rel} = a_1 - a_2$
If both bodies have equal accelerations (even while moving in opposite directions), relative acceleration is zero. Velocities do not affect relative acceleration.

27. **Correct answer: A**
Explanation: Speed can increase even with negative acceleration if motion is in the negative direction.
Sign of acceleration depends on chosen direction: $a = \frac{dv}{dt}$
Negative sign does not necessarily mean decrease in speed.

28. **Correct answer: D**
Explanation: Slope of displacement–time graph gives velocity: $v = \frac{ds}{dt}$
A line perpendicular to time axis has infinite slope ⇒ velocity undefined, not zero.
Reason is true, assertion is false.

29. **Correct answer: A**

30. **Correct answer:**

(a) → R) Zero acceleration
(b) → Q) Variable velocity
(c) → P) Displacement
(d) → S) Acceleration

Explanation:

- Straight-line v–t graph parallel to time axis ⇒ velocity constant ⇒ $a = \frac{dv}{dt} = 0$
- Curved s–t graph ⇒ slope (velocity) changes with time ⇒ variable velocity
- Area under v–t graph gives displacement: $s = \int v\, dt$
- Slope of v–t graph gives acceleration: $a = \frac{dv}{dt}$

31. Correct answer:

(a) → P) Remains constant
(b) → S) Changes continuously
(c) → R) Always towards the centre
(d) → Q) Non-zero despite constant speed

Explanation: In uniform circular motion, speed remains constant.
Velocity changes continuously due to continuous change in direction.
Direction of acceleration is always towards the centre (centripetal).
Acceleration is non-zero even though speed is constant, since velocity direction changes.

32. Correct answer: D) 0.5 m/s²

Explanation: $a = \frac{v-u}{t} = \frac{0-10}{20} = -0.5$ m/s^2
Retardation is defined as negative acceleration or the magnitude of deceleration. Since acceleration is -0.5m/s^2, the retardation is simply the positive value of this deceleration.

$$(\text{Retardation} = -\text{a} = 0.5\,\text{m/s}^2)$$

33. Correct answer: B) s/m

Explanation:
Speed unit = distance/time. s/m = $\frac{\text{time}}{\text{distance}}$ ⇒ not a speed unit

34. Correct answer: C) km/s

Explanation:
Acceleration unit = velocity/time = distance/time². km/s = $\frac{\text{distance}}{\text{time}}$ ⇒ unit of speed, not acceleration

35. Correct answer: C) 500 m

Explanation: Average velocity = 72 km/h = 20 m/s

$$\text{Distance} = v_{avg} \times t = 20 \times 25 = 500\ \text{m}$$

36. Correct answer: B) 25 m s^{-1}

Explanation: Let acceleration time = t_1.

$$v_{max} = at_1 = 5t_1$$

Total time: $2t_1 + t_2 = 25$
Distance (area under v – t graph):

$$500 = v_{max}(t_1 + t_2) = 5t_1(25 - t_1) \Rightarrow t_1 = 5\ \text{s}$$

$$v_{max} = 5 \times 5 = 25\ \text{m/s}$$

37. Correct answer: Q37: D) 15 s

Explanation: $t_2 = 25 - 2t_1 = 25 - 10 = 15$ s

Car moves with uniform speed during t_2.

38. Correct answer: 2 m/s²

Explanation: Acceleration is slope of v–t graph:

$$a = \frac{v-u}{t} = \frac{25-5}{10} = 2\ \text{m/s}^2$$

39. Correct answer: 80 m

Explanation: Displacement = area under v–t graph (triangle):

$$s = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 20 = 80\ \text{m}$$

40. Correct answer: Infinite times

Explanation:
In uniform circular motion, velocity direction changes continuously at every instant due to centripetal acceleration.

$$a = \frac{v^2}{r}\ \text{(direction always changing)}$$

2. Force and laws of motion

Solutions

Worksheet – 1

1. **Correct answer: (B) Some external force acts on it**
   Explanation: By Newton's 1st law, if net external force ΣF = 0, body stays at rest/uniform motion; it changes only when an external unbalanced force acts.
2. **Correct answer: (A) Greater mass → greater inertia**
   Explanation: In translatory motion, inertia ∝ mass (I ∝ m); also from F = ma, for same F larger m gives smaller a → more resistance to change.
3. **Correct answer: (C) Resultant force is zero (balanced forces)**
   Explanation: Rope doesn't move because forces cancel: ΣF = 0 ⇒ a = 0 (Newton's 1st/2nd law).
4. **Correct answer: (B) Inertia of direction**
   Explanation: When string breaks, centripetal force becomes zero, so the stone continues in straight line tangent due to inertia: $F_c = mv^2/r$ disappears.
5. **Correct answer: (C) Gravitational force**
   Explanation: Gravity between masses is always attractive: $F = G(m_1m_2)/r^2$.
6. **Correct answer: (A) -20 N**
   Explanation: Spring force is restoring: F = -kx = -(200)(0.10) = -20 N.
7. **Correct answer: (B) Double**
   Explanation: F = kx → if x doubles, F doubles (k constant).
8. **Correct answer: (A) -10 N**
   Explanation: F = -kx = -(500)(0.02) = -10 N.
9. **Correct answer: (A) Inertia of motion**
   Explanation: Body tends to keep moving forward when train stops; ΣF stops train but upper body continues due to inertia.
10. **Correct answer: (B) Balanced forces changing shape**
    Explanation: Equal opposite forces give ΣF = 0 (no motion) but can deform the object (shape change).
11. **Correct answer: Forces = 50 N and 30 N in opposite directions**
    Net force = 50 − 30 = 20 N, toward the larger force (50 N).
    Correct answer: (B) 20 N, in the direction of 50 N force
    Explanation: Opposite forces subtract; direction follows the bigger force.
12. **Correct answer: Forces shown = 55 N to the right, and 35 N + 20 N to the left**
    Left total = 35 + 20 = 55 N
    Right total = 55 N
    Net force = 55 − 55 = 0 N Correct answer: (D) 0 N, No direction (Forces are balanced)
    Explanation: Equal forces in opposite directions cancel each other.

Worksheet-2

1. **Correct answer: (A) Force is an action which produces acceleration**
   Explanation: When pushed, the ball changes its velocity from 0 → some value.
   So acceleration is produced due to the applied force.
   By Newton's 2nd law: $F = ma$, if $F \neq 0$then $a \neq 0$.
2. **Correct answer: (B) Impulsive force**
   Explanation: A force acting for a short time causing a finite change in momentum is impulse.
   Impulse: $J = F\Delta t = \Delta p$.
   This exactly matches the statement.

3. **Correct answer: (B) -10 N**
Explanation: $m = 5\ kg,\ u = 10\ m/s,\ v = 4\ m/s,\ t = 3\ s$
Acceleration: $a = \frac{v-u}{t} = \frac{4-10}{3} = -2\ m/s^2$
Force: $F = ma = 5 \times (-2) = -10\ N$
4. **Correct answer: (B) It has huge mass**
Explanation: Damage depends on momentum/force during collision.
Momentum: $p = mv$, even if $v$is low, large $m$gives large p.
Also kinetic energy: $KE = \frac{1}{2}mv^2$increases with mass.
5. **Correct answer: (A) Increase time of stopping and reduce force**
Explanation: Change in momentum is same, but increasing stopping time reduces force. Impulse: $F\Delta t = \Delta p$
So $F = \frac{\Delta p}{\Delta t}$becomes smaller when Δtincreases.
6. **Correct answer: (A) Time of impact is low, so force becomes large**
Explanation: Breaking needs very large force in a short time.
Impulse relation: $F = \frac{\Delta p}{\Delta t}$
When $\Delta t$is very small, Fbecomes very large.
7. **Correct answer: (C) 8 N**
Explanation: $m = 4\ kg,\ u = 3\ m/s,\ v = 13\ m/s,\ t = 5\ s$
Acceleration: $a = \frac{13-3}{5} = 2\ m/s^2$
Force: $F = ma = 4 \times 2 = 8\ N$
8. **Correct answer: (B) 5 m/s**
Explanation: Given $F = 20\ N,\ t = 0.5\ s,\ m = 2\ kg,\ u = 0$
Impulse: $\Delta p = F\Delta t = 20 \times 0.5 = 10\ kg\ m/s$
So $mv = 10 \Rightarrow v = \frac{10}{2} = 5\ m/s$
9. **Correct answer: (C) 25 N**
Explanation: $m = 5\ kg,\ u = -\frac{4\ m}{s},\ v = +\frac{6\ m}{s},\ t = 2\ s$
Acceleration: $a = \frac{6-(-4)}{2} = \frac{10}{2} = \frac{5\ m}{s^2}$
Force magnitude: $|F| = ma = 5 \times 5 = 25\ N$
10. **Correct answer: (B) 2 s**
Explanation: $m = 4\ kg,\ u = 7\ m/s,\ v = 0,\ F = -14\ N$
Acceleration: $a = \frac{F}{m} = \frac{-14}{4} = -3.5\ m/s^2$
Time: $t = \frac{v-u}{a} = \frac{0-7}{-3.5} = 2\ s$
11. **Correct answer: Impulse = Force × time**
= 200 × 0.02 = 4 N·s
Answer: (B) 4 N·s
12. **Correct answer:From Newton's second law:** $a \propto F$when mass is constant
Answer: (B) Force
13. **Correct answer:Impulse = change in momentum = m(v − u)**
= 3 × (5 − 1) = 3 × 4 = 12 N·s
Answer: (C) 12 N·s
14. **Correct answer:From $F = ma$: when acceleration is constant,** $F \propto m$
Answer: (C) Mass

Work sheet-3

1. **Correct answer:(C) Exert an equal force on A in opposite direction instantly**
Explanation: Newton's 3rd law: every action has an equal and opposite reaction.
So if A exerts force on B, B exerts equal magnitude force on A in opposite direction.
Formula idea: $F_{AB} = -F_{BA}$ (same time).
2. **Correct answer: (B) Because they act on different bodies**
Explanation: Action and reaction are equal and opposite but act on two different bodies.
Balanced forces must act on the same body to cancel.
So they never cancel each other on one object.
3. **Correct answer: (A) 4 kg·m/s and 1 m/s**
Explanation: Bullet momentum: $p = mv = 0.02 \times 200 = 4\ kg \cdot m/s$
Initial momentum = 0, so gun momentum = -4.
Recoil speed: $v_g = \frac{p}{M} = \frac{4}{4} = 1\ m/s$(backward).

4. **Correct answer: (B) Internal forces of the system**

   Explanation: In a two-ball system, the force of A on B and B on A are mutual interaction forces.
   They act within the system, so they are internal.
   They cancel in total momentum calculation.

5. **Correct answer: (C) Total momentum must remain conserved**

   Explanation: Initially gun + bullet system momentum = 0.
   After firing, bullet gets forward momentum, so gun must get equal backward momentum.
   Conservation: $mv = MV$ (opposite directions).

6. **Correct answer: (B) Action and reaction occur as equal and opposite forces**

   Explanation: Boy pushes boat backward → boat pushes boy forward with equal force.
   This is Newton's 3rd law action-reaction pair.
   So boy moves forward, boat moves backward.

7. **Correct answer: (B) -p**

   Explanation: No external force ⇒ total momentum remains constant.
   Initial total momentum = 0, so final must also be 0.
   If one gets $+p$, the other must be $-p$.

8. **Correct answer: (C) 18 kg·m/s**

   Explanation: Total momentum: $p = m_1u_1 + m_2u_2$
   $= 2 \times 6 + 3 \times 2 = 12 + 6 = 18\ kg \cdot m/s$.
   Hence option (C).

9. **Correct answer: (B) -3 m/s**

   Explanation: Initial momentum = 0. Final: $mv + MV = 0$.
   So recoil velocity
   y: $V = -\frac{mv}{M} = -\frac{0.02\times300}{2} = -3\ m/s$
   Negative sign shows backward direction.

10. **Correct answer: (C) 40 N**

    Explanation: Average force: $F = \frac{\Delta p}{\Delta t} = \frac{m(v-u)}{t}$
    Here $m = 2$, $u = 8$, $v = 0$, $t = 0.4$
    $F = \frac{2(0-8)}{0.4} = -40\ N \Rightarrow$ magnitude 40 N.

Worksheet-4

1. **Correct answer: (C) 1.4 m/s²**
   Explanation: Atwood machine → $a = \frac{(4-3)g}{4+3} = \frac{1\times9.8}{7} = 1.4$ m/s^2
2. **Correct answer: (C) N = mg**
   Explanation: Block at rest on horizontal surface → vertical equilibrium → $N = mg$
3. **Correct answer: (A) 5 m/s²**
   Explanation: Total mass = 3 + 2 = 5 kg → $a = \frac{F}{M} = \frac{25}{5} = 5$ m/s^2
4. **Correct answer: (C) $f_2 = M_3F$ / total mass**
   30 = 10F / 20
   F = 60 N
5. **Correct answer: (D) 90°**
   Explanation: By definition, normal reaction is perpendicular to surface → angle = 90°.
6. **Correct answer: (D) 80 N**
   Explanation: At rest → $N = mg = 8 \times 10 = 80$ N.
7. **Correct answer: (D) 73.5 N**
   Explanation: Atwood machine → $T = \frac{2m_1m_2g}{m_1+m_2} = \frac{2\times6\times10\times9.8}{6+10} = 73.5$ N.
8. **Correct answer: (B) 120 N**
   Explanation: Vertical equilibrium → $N = mg = 12 \times 10 = 120$ N.
9. **Correct answer: (B) 2 m/s²**
   Explanation: Total mass = $4 + 6 = 10$kg → $a = \frac{F}{M} = \frac{20}{10} = 2$ m/s^2.

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (C) Upper part of the body continues to be in the state of motion, whereas the lower part of the body in contact with the seat comes to rest**
   Explanation: Due to inertia of motion, the upper body tends to keep moving

forward when train stops suddenly. Lower part in contact with seat stops with train first, causing a forward jerk. Newton's 1st law: body maintains its state unless external force acts.

2. **Correct answer: (C) Newton's first law of motion**
   Explanation: When carpet is beaten, carpet moves suddenly but dust tends to remain at rest due to inertia.
   So dust separates and falls out.
   Newton's 1st law (inertia of rest) explains it.
3. **Correct answer: (D)** 2×10^7 **N/m**^{2}
   Explanation: $\Delta p = 2mv = 2(10^{-3})(100) = 0.2$per ball.
   Force $F = n\Delta p = 10^4 \times 0.2 = 2000\ N$.
   Area $A = 1\ cm^2 = 10^{-4}m^2$, so $P = \frac{F}{A} = \frac{2000}{10^{-4}} = 2 \times 10^7$.
4. **Correct answer: (A) 5000 N**
   Explanation: $m = 100\ kg, u = 5\ m/s, v = 0, t = \frac{1}{10} = 0.1s$.
   Acceleration $a = \frac{v-u}{t} = \frac{0-5}{0.1} = -50\ m/s^2$.
   Force $F = ma = 100 \times (-50) = -5000\ N \Rightarrow$ magnitude **5000 N** opposite direction.
5. **Correct answer: (B) 2 mnu**
   Explanation: Each second $n$balls hit, each has momentum change $\Delta p = 2mu$(elastic rebound).
   Total force $F = \frac{\Delta p_{\text{total}}}{t} = n(2mu)$.
   So force on surface $= 2mnu$.
6. **Correct answer: D)** $\sqrt{2}m_1 : m_2$
   Explanation: Velocity on reaching the ground is $v = \sqrt{2gh}$.
   Momentum $p = mv$. Therefore, $p_1 = m_1\sqrt{2g(2h)}$and$p_2 = m_2\sqrt{2gh}$.
   The ratio $\frac{p_1}{p_2} = \frac{m_1\sqrt{4gh}}{m_2\sqrt{2gh}} = \frac{\sqrt{2}m_1}{m_2}$.
7. **Correct answer: C)** $\frac{a_1a_2}{a_1+a_2}$
   Explanation: Since $F = ma$, then $m_1 = \frac{F}{a_1}$and$m_2 = \frac{F}{a_2}$.
   For the combination, $a = \frac{F}{m_1+m_2} = \frac{F}{\frac{F}{a_1}+\frac{F}{a_2}}$.
   Simplifying gives $a = \frac{a_1a_2}{a_1+a_2}$.
8. **Correct answer: A) 4 m/s**
   Explanation: By Law of Conservation of Momentum, $m_bv_b + m_gv_g = 0$.
   So, $|\ v_g\ | = \frac{m_bv_b}{m_g} = \frac{0.02\ \text{kg} \times 200\ \text{m/s}}{1\ \text{kg}} = 4\ \text{m/s}$.
9. **Correct answer: D)** $\mathbf{2\ m/s^2}$
   Explanation: Using the formula derived in Question 7, $a = \frac{a_1a_2}{a_1+a_2} = \frac{6\times3}{6+3} = \frac{18}{9} = 2\ \text{m/s}^2$.
10. **Correct answer: C) 50 kg m/s**
    Explanation: Momentum $p = mv$.
    Plugging in the values, $p = 10\ \text{kg} \times 5\ \text{m/s} = 50\ \text{kg m/s}$.
11. **Correct answer: B) 100 kg m/s**
    Explanation: First find final velocity using $v = u + at$.
    Since it starts from rest, $v = 0 + (4 \times 5) = 20$ m/s.
    Then, $p = mv = 5\ \text{kg} \times 20\ \text{m/s} = 100\ \text{kg m/s}$.
12. **Correct answer: A) Inertia of rest**
    Explanation: Inertia of rest is the tendency of a body to remain in its state of rest.
    When the car starts suddenly, the lower body moves forward with the car, but the upper body remains stationary due to inertia, causing the person to fall back.
13. **Correct answer: B) Force**
    Explanation: Newton's Second Law states that force is equal to the rate of change of momentum $F = \frac{dp}{dt} = ma$.
    It provides a quantitative measure for force.
14. **Correct answer: (B) 5 m/s²**
    Explanation: From $F = ma \Rightarrow m = 12/3 = 4$kg; new $a = 20/4 = 5\ \text{m/s}^2$
15. **Correct answer: B) 2 N**
    Explanation: Using the formula
    $$F = ma,$$
    substitute the given values, $F = 10\ \text{kg} \times 0.2\ \text{m/s}^2 = 2\ \text{N}$.

16. **Correct answer: A) 1800 N**
Explanation: First, find the acceleration using
$$v^2 - u^2 = 2as.$$
Since $v = 0$(rest), $0 - 10^2 = 2(a)(50)$, so $a = -1$ m/s^2.
Then, calculate force magnitude, $F = ma = 1800 \times 1 = 1800$ N.

17. **Correct answer: B) 10 N**
Explanation: First, find acceleration using $s = ut + \frac{1}{2}at^2$.
Assuming it starts from rest ($u = 0$),
$10 = 0 + \frac{1}{2}(a)(2^2)$,
which simplifies to $10 = 2a$,
so $a = 5$ m/s^2.
Then, calculate force, $F = ma = 2 \times 5 = 10$ N.

18. **Correct answer: (D) (A)**
Velocity (matches Option A): $\vec{v} = \vec{u} + \vec{a}t = 0 + (1000\hat{\imath} + 500\hat{\jmath})(12.5) = 12500\hat{\imath} + 6250\hat{\jmath}$ m/s.
Actual Position: $\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2 = \frac{1}{2}(1000\hat{\imath} + 500\hat{\jmath})(12.5)^2 = 78125\hat{\imath} + 39062.5\hat{\jmath}$ m.

19. **Correct answer: (B) 6250 N**
Explanation: Given $u = 250\, m/s$, $v = 0$, $s = 5cm = 0.05m$.
Use $v^2 = u^2 + 2as \Rightarrow a = \frac{-u^2}{2s} = \frac{-250^2}{0.1} = -625000\, m/s^2$.
Force magnitude $F = ma = 0.01 \times 625000 = 6250\, N$.

20. **Correct answer: (D) 180°**
Explanation: Action of book on table is downward force, reaction of table on book is upward normal.
They are equal and opposite (Newton's 3rd law).
Angle between opposite vectors = 180°.

21. **Correct answer: (C) 28 N·s**
Explanation: Impulse = change in momentum = $p_f - p_i$.
Initial momentum = $4 \times 3 = 12$; final = $40 \rightarrow$ Impulse = $40 - 12 = 28$N·s

22. **Correct answer: (C) 15 N·s, 1500 N**
Explanation: Impulse = $mv = 0.05 \times 300 = 15$N·s;
Force = Impulse/$t = 15/0.01 = 1500$N

23. **Correct answer: (C) 40 N**
Explanation: Using $F = ma$. Same body $\Rightarrow$ mass constant.
$m = 10/2 = 5$kg $\rightarrow$ New force $F = 5 \times 8 = 40$N

24. **Correct answer: (B) 2.5 N**
Explanation: Smooth surface $\rightarrow$ common acceleration
$a = \frac{F}{m_1+m_2} = \frac{6}{7+5} = 0.5$ m/s^2.
Force on 5 kg block = contact force = $5 \times 0.5 = 2.5$ N.

25. **Correct answer: (C) g/3**
Explanation: Atwood machine $\rightarrow a = \frac{(10-5)g}{10+5} = \frac{5g}{15} = \frac{g}{3}$.

26. **Correct answer: (C) 8 m/s²**
Explanation: Movable pulley pulled by force $\rightarrow$ two string segments pull block $\rightarrow 2T = F \Rightarrow T = 96/2 = 48$ N.
Acceleration $a = \frac{T}{m} = \frac{48}{6} = 8$ m/s^2.

27. **Correct answer: (C) 32 m/s**
Explanation: Same setup $\rightarrow 2T = 32 \Rightarrow T = 16$ N.
$a = \frac{T}{m} = \frac{16}{2} = 8$ m/s^2. From rest $\rightarrow v = at = 8 \times 4 = 32$ m/s.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

28. **Correct answer: (A), (B), (C)**
Explanation: Momentum $p = mv$is "quantity of motion" $\Rightarrow$ **(C)** correct.
Units are $kg\, m/s$and $g\, cm/s \Rightarrow$ **(B)** correct.
Also $F = \frac{\Delta p}{\Delta t}$; to stop in unit time, required force $F = p \Rightarrow$ **(A)** correct.

29. **Correct answer: (A) Newton, (B) Dyne, (C) kg-f, (D) gm-f**
Explanation: Force units include SI Newton (N) and CGS Dyne.
Also gravitational units kgf and gf are used (non-SI).
So all given are units of force.

30. **Correct answer: (A), (B), (D)**
Explanation: Gun recoils when bullet moves forward $\Rightarrow$ action–reaction pair $\Rightarrow$ (A) true.
Swimming works by pushing water backward $\Rightarrow$ reaction pushes swimmer

forward ⇒ (B) true.
Forces always occur in pairs ⇒ (D) true.
(C) false because action & reaction act on different bodies.

31. Correct answer: (A) $T_1 = 25\ N$ **and (D)** $T_2 = 25\sqrt{3}\ N$
Explanation: For equilibrium, resolve tensions: Horizontal: $T_1 \cos\ 30° = T_2 \cos\ 60° \Rightarrow T_2 = \sqrt{3}T_1$.
Vertical: $T_1 \sin\ 30° + T_2 \sin\ 60° = mg = 50$.
Substitute ⇒ $\frac{T_1}{2} + \frac{3T_1}{2} = 50 \Rightarrow T_1 = 25\ N$, $T_2 = 25\sqrt{3}\ N$.

ASSERTION AND REASNING TYPE

32. Correct answer: (C) A is true, but R is false
Explanation: Balanced forces give net force = 0, so they cannot change motion (no acceleration).
But they can deform/change shape (e.g., squeezing).
Reason is false because balanced forces do not always act same direction and do not increase speed.
Formula: $F_{\text{net}} = 0 \Rightarrow a = 0$.

33. Correct answer: (B) Both A and R are true, but R is NOT the correct explanation of A
Explanation: Assertion true: momentum $p = mv$.
Reason is also true because $F = \frac{\Delta p}{\Delta t} \Rightarrow p = F \cdot 1s$(force to stop in unit time).
But R is not the fundamental explanation of $p = mv$; it's an alternate interpretation.
Formula: $p = mv$, $F = \frac{\Delta p}{\Delta t}$.

34. Correct answer: (A) Both A and R are true, and R is the correct explanation of A
Explanation: If no external force, total momentum remains conserved in collision.
Internal forces are equal and opposite (Newton's 3rd law) so net internal impulse cancels.
Hence momentum before = momentum after.
Formula: $\sum F_{\text{ext}} = 0 \Rightarrow \Delta P_{\text{total}} = 0$.

35. Correct answer: (B) Both A and R are true, but R is NOT the correct explanation of A
Explanation: Assertion true: bullet pushes gun backward with equal and opposite force.
Reason true: initially total momentum is zero since system at rest.
But recoil happens mainly due to action–reaction during firing; R supports conservation, not directly A.
Formula: $F_{bg} = -F_{gb}$and $P_i = 0 = P_f$.

MATCH THE FOLLOWING TYPE

36. Correct answer:
a−S, b−R, c−Q, d−P
Explanation:

A. Inertia – (S) Measurement of mass
Mass is the quantitative measure of an object's inertia (resistance to change in motion).

B. Linear momentum – (R) Quantity of motion contained in the body
Newton defined momentum ($p = mv$) as the "quantity of motion."

C. Newton's Second Law – (Q) Rate of change of momentum
The law states that force is directly proportional to the rate of change of momentum $F = \frac{\Delta p}{\Delta t}$.

D. Newton's Third Law – (P) Action = – reaction
For every action force, there is an equal and opposite reaction force.

37. Correct answer: a−R, b−R, c−P, d−Q
Explanation: Given initial momentum, $p = mv = 24$ kg m/s.

A. Twice mass ($2m$), same speed (v)
$$p' = (2m)(v) = 2(mv) = 2 \times 24 = \mathbf{48}\ \text{kg m/s}$$
(Matches R)

B. Same mass (m), twice speed ($2v$)
$$p' = (m)(2v) = 2(mv) = 2 \times 24 = \mathbf{48}\ \text{kg m/s}$$
(Matches R)

C. Half mass ($0.5m$), twice speed ($2v$)
$$p' = (0.5m)(2v) = 1(mv) = \mathbf{24}\ \text{kg m/s}$$
(Matches P)

D. Same mass (m), half speed ($0.5v$)

$$p' = (m)(0.5v) = 0.5(mv)$$
$$= 0.5 \times 24$$
$$= \mathbf{12}\ \text{kg m/s}$$

(Matches Q)

COMPREHENSHION TYPE

38. Correct answer: (B) 0.5 kg

Explanation: Given same force $F = 5N$ produces $a_1 = 10\ m/s^2$. $m_1 = \frac{F}{a_1} = \frac{5}{10} = 0.5\ kg$

Using $F = ma$.

39. Correct answer: (C) 0.25 kg

Explanation: Given same force $F = 5N$ produces $a_2 = 20\ m/s^2$. $m_2 = \frac{F}{a_2} = \frac{5}{20} = 0.25\ kg$

40. Correct answer: (D) 6.67 m/s²

Explanation: Combined mass $m = m_1 + m_2 = 0.5 + 0.25 = 0.75\ kg$.

Same force $F = 5N$ on combined body:

$a = \frac{F}{m} = \frac{5}{0.75} = 6.67\ m/s^2$

41. Correct answer: (A) 1.6 N

Explanation: $F = ma = 0.8 \times 2 = 1.6$ N.

42. Correct answer: (B) 2 m/s²

Explanation: $\boldsymbol{a = \frac{v-u}{t} = \frac{12-2}{5} = 2\ \text{m/s}^2}$.

43. Correct answer: 2

Explanation: $F = 100$dynes, $m = 5g$.

Acceleration $a = \frac{F}{m} = \frac{100}{5} = 20\ \text{cm/s}^2$.

Velocity in $10s$: $v = at = 20 \times 10 = 200$ cm/s $= 2 \times 100$ cm/s.

44. Correct answer: 12 N

Explanation: $m = 2\ kg, u = 18\ m/s, v = 0, t = 3s$.

Average force $F = \frac{\Delta p}{\Delta t} = \frac{m(v-u)}{t} = \frac{2(0-18)}{3} = -12N$.

Magnitude $= 12N$.

45. Correct answer: 30 kg·m/s

Explanation: Initially total momentum of boy + boat system is 0 (at rest).

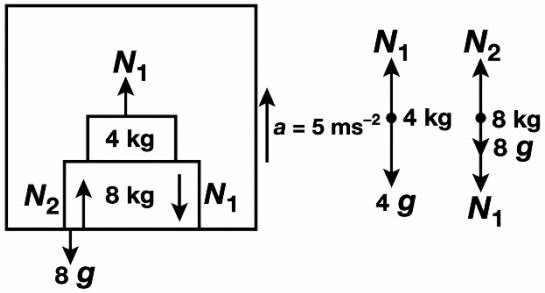
Boat gets 30 kg.backward ⇒ boy must get equal forward momentum.

Previous Year Questions

1. Correct answer: (B)

Normal forces are always perpendicular to the plane surface at the point of contact.

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For 4 kg block: $N_1 - 4g = 4a \Rightarrow N_1 = 60$ N

For 8 kg block: $N_2 - N_1 - 8g = 8a$

$N_2 = 8 \times 5 + 80 + 60$

$N_2 = 180$ N

Hence, only statement (ii) is incorrect.

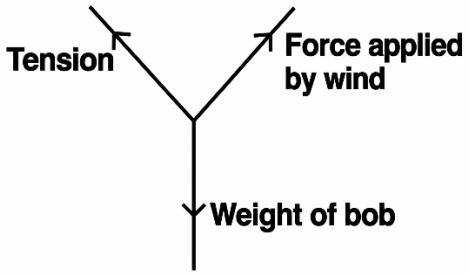
2. Correct Answer: B

Explanation: Statement 1 is True: On a rainy day, water acts as a lubricant between the tires and the road. This significantly reduces friction, making it difficult to maintain control or stop the vehicle at high speeds.

Statement 2 is True: According to the laws of dry friction, the frictional force depends on the nature of the surfaces and the normal force, but it is independent of the apparent area of contact.

3. Correct answer: (C)

The figure given in option (C) correctly represents the forces acting on the pendulum bob.



4. Correct answer: (D)

Mass of helicopter, $M = 2000$ kg
Acceleration, $a = 15$ m s^{-2}
Total mass of crew and passengers, $m = 500$ kg

$$g = 10 \text{ m s}^{-2}$$

(i) Force on floor by crew and passengers:
$F = m(g + a) = 500(10 + 15)$
$F = 1.25 \times 10^4$ N (downward)
(ii) Action of rotor: $(M + m)(g + a) = (2000 + 500)(10 + 15)$
$= 6.25 \times 10^4$ N (downward)
(iii) Force on helicopter due to air:
6.25×10^4 N (upward)

5. Correct answer: (A)

In region AB, the x-t graph is a straight line.
Hence, velocity is constant, acceleration is zero, and net force is zero in this region.

6. Correct answer: (C)

Definition of force is given by Newton's first law.
Measure of force is given by Newton's second law.
Effect of force is given by impulse.
Recoiling of a gun is explained by Newton's third law.

Matching: $P \rightarrow (iv)$, $Q \rightarrow (iii)$, $R \rightarrow (ii)$, $S \rightarrow (i)$

7. Correct answer: (A)

$F_A = F_B$
$M_A a_A = M_B a_B$
$M_A a_A = M_B (3a_A)$
$M_A = 3M_B$
$\therefore$ Mass of A is three times that of B.

8. Correct answer: (A)

Initial momentum of the body $= Mu$
Final momentum of the body $= -Mu$ (negative sign due to change in direction)

$\therefore$ Change in momentum
$= Mu - (-Mu) = 2Mu$

3. GRAVITATION SOLUTIONS

Solutions

Worksheet – 1

1. **Correct answer: (D) None of these**

   Explanation: Gravitational constant G is universal and independent of mass, size, or nature of bodies.
   Formula: $F = G\frac{m_1 m_2}{r^2}$

2. **Correct answer: (A) Mass × acceleration due to gravity**

   Explanation: Weight $W = mg$.
   Formula: $W = m \times g$

3. **Correct answer: (D) Anywhere in the universe**

   Explanation: Newton's law of gravitation is universal.
   Formula: $F = G\frac{m_1 m_2}{r^2}$

4. **Correct answer: (D) Both (A) and (C)**

   Explanation: Force is directly proportional to product of masses and inversely proportional to square of distance.
   Formula: $F \propto \frac{m_1 m_2}{r^2}$

5. **Correct answer: (C) 6.4 × 10^6 m**

   Reason: Radius of Earth ≈ 6400 km = 6.4 × 10^6 m.

6. **Correct answer: (B) 6.6 × 10^{-8} dyne cm^2 g^{-2}**

   Reason: CGS value of G = 6.67 × 10^{-8} dyne cm^2 g^{-2}.

7. **Correct answer: (C) Become double**

   Explanation: $g = \frac{GM}{R^2}$; if mass doubles, g doubles.
   Formula: $g \propto M$

8. **Correct answer: (C) Independent of mass of the body**

   Reason: g depends only on Earth's mass and radius.

9. **Correct answer: (C) 25 m/s**

   Using v = u + gt = 5 + (10×2) = 25 m/s.

10. **Correct answer: (C) 20 m**

    Using s = ut + ½gt^2 = 0 + ½(10)(2^2) = 20 m.

Worksheet – 2

1. **Correct answer: (C) g = GM/R^2**

   Explanation: From Newton's law, $g = \frac{GM}{R^2}$.

2. **Correct answer: (C) g_p = GM/R_p^2**

   Explanation: From Newton's law, $g = \frac{GM}{R^2}$.

3. **Correct answer: (C) decreases**

   Explanation: $g = \frac{GM}{(R+h)^2}$, increases in height ⇒ g decreases.

4. **Correct answer: (B) (2h/R) × 100%**

   Reason: Percentage decrease = (2h/R) ×100%.

5. **Correct answer: (C) Distance from centre of Earth**

   Reason: g depends on r from Earth's centre.

6. **Correct answer: (B) Centrifugal force changes**

   Reason: Rotation of Earth causes change in centrifugal force with latitude.

7. **Correct answer: (C) Decreases linearly**

   Reason: g_n = g(1 − d/R).

8. **Correct answer: (B) Decreases**

   Reason: g decreases as distance from Earth increases.

9. **Correct answer: (A) 9.8 m/s^2**

   Explanation:
   Planet mass = 4M, radius = 2R

$$g' = \frac{G(4M)}{(2R)^2} = \frac{4GM}{4R^2} = g$$

10. Correct answer: (D) 1.59 × 10^6 m

At height h:

$$g_h = g\left(\frac{R}{R+h}\right)^2$$

At depth d:

$$g_d = g\left(1-\frac{d}{R}\right)$$

Given $g_d = \frac{1}{2}g_h$.

$$1-\frac{d}{R} = \frac{1}{2}\left(\frac{R}{R+h}\right)^2$$

With $R = 6400$km and $h = 1600$km:

$$\frac{R}{R+h} = \frac{6400}{8000} = 0.8$$
$$1-\frac{d}{R} = \frac{1}{2}(0.8)^2 = 0.32$$
$$\frac{d}{R} = 0.68$$
$$d = 0.68R = 0.68 \times 6400 = 4352 \text{ km}$$

Worksheet – 3

1. Correct answer: (C) Explanation: The escape velocity formula is

$$v_e = \sqrt{\frac{2GM}{R}}$$

where $M$is the mass of the planet. It is independent of the mass of the escaping object.

2. Correct answer: (D) 2.83 years

Explanation: Using Kepler's Third Law,

$$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{R_2}{R_1}\right)^3$$

If $R_2 = 2R_1$, then

$T_2 = 1$ year $\times \sqrt{2^3} = \sqrt{8} \approx 2.828$ years.

3. Correct answer: (B)

Explanation: Kepler's First Law (Law of Orbits) states that planetary orbits are elliptical with the Sun at one focus.

4. Correct answer: (A) $T^2 \propto R^3$

Explanation: Kepler's Third Law (Law of Periods) states that the square of the time period is proportional to the cube of the semi-major axis:

$$T^2 = \left(\frac{4\pi^2}{GM}\right)R^3.$$

5. Correct answer: (B)

Explanation: These are the primary practical applications of artificial satellites in orbit around Earth.

6. Correct answer: (C) Geostationary satellite

Explanation: These satellites have an orbital period of 24 hours, matching Earth's rotation, making them appear fixed over a specific longitudinal point.

7. Correct answer: (B) 7.9 km/s

Explanation: Using

$$v_0 = \sqrt{gR}$$

with $g \approx 9.8$ m/s^{2}and $R \approx 6.4 \times 10^6$ m, the result is ≈ 7919 m/sor 7.9 km/s.

8. Correct answer: (A) Half of original

Explanation: Orbital velocity

$$v \propto \frac{1}{\sqrt{r}}.$$

If $r$becomes $4r$, the velocity becomes $\frac{1}{\sqrt{4}} = \frac{1}{2}$of the original.

9. Correct answer: (A) 3.95 km/s

Explanation: Since

$$v \propto \frac{1}{\sqrt{r}},$$

if the distance increases by 4 times, the velocity decreases by a factor of $\sqrt{4} = 2$. So, $7.9/2 = 3.95$ km/s.

10. Correct answer: (B) v/3

Explanation: Based on

$$v = \sqrt{\frac{GM}{r}},$$

if $r$is replaced by $9r$, the new speed

$$v' = \sqrt{\frac{GM}{9r}} = \frac{1}{3}\sqrt{\frac{GM}{r}} = \frac{v}{3}.$$

Worksheet – 4

1. **Correct answer: (C) $v_e = \sqrt{2}\, v_o$**

   Explanation: Escape velocity $v_e = \sqrt{\frac{2GM}{R}}$ and orbital velocity $v_o = \sqrt{\frac{GM}{R}}$. Dividing gives $\frac{v_e}{v_o} = \sqrt{2}$.

2. **Correct answer: (A)**

   Reason: $h = \left(\frac{T^2 g R^2}{4\pi^2}\right)^{\frac{1}{3}} - R$

3. **Correct answer: (C)**

   Reason: Both gravitational force and reaction by surface act on body.

4. **Correct answer: (C)**

   Reason: Equation is $GMm/r^2 - R = ma$.

5. **Correct answer: (B) Creation of artificial gravity**

   Reason: Artificial gravity can be produced by rotating the satellite, creating centrifugal force.

6. **Correct answer: (C)**

   Reason: In free fall, no normal reaction acts, so apparent weight is zero.

7. **Correct answer: (B) –GMm/r**

   Reason: Gravitational potential energy is defined relative to infinity and is negative.

8. **Correct answer: (B) GMm/2r**

   Reason: KE=21mvo2=21m(rGM)=2rGMm

9. **Correct answer: (C)**

   Reason: Geostationary orbit must be equatorial and concentric.

10. **Correct answer: (C) Central**

    Reason: Gravitational force acts along radius → torque = 0.

COMPETETIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct answer: (B)**

   Mass is an intrinsic property and remains constant. Weight changes because it depends on local gravity: $W = mg$.

2. **Correct answer: (A)**

   Explanation: This is Kepler's Second Law (Law of Areas), which states that the areal velocity of a planet remains constant.

3. **Correct answer: (C)**

   Reason: Kinetic energy is minimum when planet is farthest from the Sun (aphelion).

4. **Correct answer: (C)**

   Explanation: By conservation of angular momentum ($L = mvr$), when distance decreases at perihelion, velocity increases.

5. **Correct answer: (B) Strong nuclear force**

   Reason: Strong nuclear force binds protons and neutrons in nucleus.

6. **Correct answer: (D) $5^{3/2}$ years**

   Reason: $T \propto R^{3/2} \Rightarrow T = 5^{3/2}$ years.

7. **Correct answer: (C)**

   A satellite moves in a circular orbit, so it needs a centripetal acceleration toward the center of the Earth. This acceleration is provided by Earth's gravitational force.

   From Newton's law of gravitation, the acceleration due to gravity at distance *r* from Earth's center is:

   $$a = \frac{GM}{r^2}$$

8. **Correct answer: (A)**

   Gravitational force is always attractive; there is no repulsive gravitational interaction.

9. **Correct answer: (B) Gravity**

   Reason: Atmosphere is held by Earth's gravitational pull.

10. **Correct answer: (A)**

    Explanation: Effective gravity at equator is $g' = g - \omega^2 R$.
    If Earth stops rotating ($\omega = 0$), centrifugal effect vanishes and g'increases.

11. Correct answer: (D) Increase by 2%

Explanation: Since $g = \frac{GM}{R^2}$, if mass $M$is constant, then $g \propto R^{-2}$. For small changes,
$\frac{\Delta g}{g} = -2\frac{\Delta R}{R}$.
If Rdecreases by 1%, $g$increases by 2%.

12. Correct answer: (A)

Explanation: The acceleration due to gravity inside the earth is given by
$g_d = g\left(1-\frac{d}{R}\right)$.
As depth $d$increases toward the center, $g$decreases linearly and becomes zero at the center.

13. Correct answer: (C) g(1 − 2h/R)

Explanation: The formula at height is
$g_h = g\left(1+\frac{h}{R}\right)^{-2}$.
For $h \ll R$, using binomial expansion,
$g_h \approx g\left(1-\frac{2h}{R}\right)$.

14. Correct answer: (A) g(1 − h/R)

Explanation: Inside the earth at depth h, only the mass within radius $(R-h)$attracts the body.
Thus, $g_h = g\left(1-\frac{h}{R}\right)$.

15. Correct answer: (B) Fc = mv²/R

Explanation: Centripetal force equals mass × centripetal acceleration.
Since $a_c = \frac{v^2}{R}$,
$F_c = m\frac{v^2}{R}$.

16. Correct answer: (B)

Reason: F ∝ 1/r² ⇒ doubling distance reduces force to 1/4.

17. Correct answer: (D) F = GMm/R²

Explanation: According to the universal law of gravitation, force is proportional to the product of masses and inversely proportional to the square of distance.

18. Correct answer: (C) Gravitational force

Explanation: Gravity pulls water and objects downward toward the Earth's center, causing flow along slopes and hills.

19. Correct answer: (A) 4.17 × 10^{42}

Explanation:

$$\frac{F_e}{F_g} = \frac{ke^2}{Gm^2}$$

Substituting constants gives approximately
4.17×10^{42}.

20. Correct answer: (C) 5 × 10^{-8} N

Explanation:

$F = \frac{Gm_1m_2}{r^2} = \frac{6.67\times10^{-11}\times50\times60}{2^2} \approx 5.0 \times 10^{-8}\ N$

21. Correct answer: (A) 3%

Explanation: Acceleration due to gravity is $g = \frac{GM}{R^2}$. For small changes,
$\frac{\Delta g}{g} = -2\frac{\Delta R}{R}$.
If Rdecreases by 1.5%, then $g$increases by $2 \times 1.5\% = 3\%$.

22. Correct answer: (D) g_1:g_2 = $R_1\rho_1$: $R_2\rho_2$

Explanation: $M = \frac{4}{3}\pi R^3\rho$.
So $g = \frac{GM}{R^2} = \frac{4}{3}\pi GR\rho$.
Hence $g \propto R\rho$.

23. Correct answer: (B) 6336 km

Explanation: At depth d, $g' = g(1-\frac{d}{R})$.
Given $g' = 0.01g$:
$0.01 = 1 - \frac{d}{R} \Rightarrow \frac{d}{R} = 0.99$.
$d = 0.99 \times 6400 = 6336$km.

24. Correct answer: (C) 2624 km

Explanation: $W \propto \frac{1}{(R+h)^2}$.
For half weight:
$\frac{1}{2} = (\frac{R}{R+h})^2 \Rightarrow h = R(\sqrt{2}-1)$.
$h \approx 6400 \times 0.414 \approx 2650$km.

25. Correct answer: (B)

Explanation: Circular or elliptical motion requires a centripetal force toward the Sun. Without it, Earth would move in a straight line.

26. Correct answer: (D)

Reason: In circular motion, velocity and acceleration change continuously, speed constant.

27. Correct answer: (B) v^2/r

Reason: Formula of centripetal acceleration.

28. Correct answer: (A) 2/3 m

Explanation:

$$\frac{4}{x^2} = \frac{1}{(1-x)^2} \Rightarrow \frac{2}{x} = \frac{1}{1-x} \Rightarrow x = \frac{2}{3}\ \text{m}$$

Correct answer is $\frac{2}{3}$**m**,

29. Correct answer: (B)

Explanation: Gravitational force is a central force, so torque is zero. Hence angular momentum is conserved.

30. Correct answer: (C) 3R

Explanation:

$$\frac{1}{16} = (\frac{R}{R+h})^2 \Rightarrow \frac{1}{4} = \frac{R}{R+h} \Rightarrow R + h = 4R \Rightarrow h = 3R$$

ONE OR MORE THAN ONE CORRECT

31. Correct answer: A, B, C

Reason: G = 6.67×10^{-11} N m^2/kg^2, independent of bodies and medium; it is constant everywhere.

32. Correct answer: A, B, D

Reason: If G decreases, gravitational pull weakens → day and year increase, kinetic energy decreases; Earth will not spiral inward.

33. Correct answer: D) Increase by about 56%.

If the radius becomes 80% of its original value, then

$$R' = 0.8R$$

Mass M remains the same.

New gravity:

$$g' = \frac{GM}{(0.8R)^2} = \frac{GM}{0.64R^2} = \frac{g}{0.64} = 1.56g$$

34. Correct answer: A, B, C

Reason: Forces are action–reaction pair, G is scalar, g is vector, both are experimentally measured.

35. Correct answer: A, B

Reason: Kepler's III law: $T^2 \propto R^3$ and $(T_1/T_2)^2 = (R_1/R_2)^3$.

ASSERTION & REASON

36. Correct answer: (a)

Reason: T depends on $(R+h)^3$, so knowing T gives h. Both true and correctly related.

37. Correct answer: (a)

Reason: Gravity is independent of medium; reason explains assertion.

38. Correct answer: (a)

Reason: Downward motion takes +g, hence v = u + gt.

39. Correct answer: (a)

Reason: Constant areal velocity is equivalent to conservation of angular momentum.

MATRIX MATCHING

40. Correct answer:

A → R (Escape velocity → Minimum velocity to escape Earth)
B → Q (Mass → Determined by physical balance)
C → P (Law of Areas → Sweeps equal areas in equal intervals)
D → D (Weight → Attractive force exerted by Earth)

41. Correct answer:

(a) Escape velocity → (3) √2 times orbital velocity
(b) Time period of revolution → (1) Time for one revolution
(c) Geostationary satellite → (4) Period = Earth's rotation
(d) Weightlessness → (2) Apparent weight zero

COMPREHENSION

42. Correct answer: (C)

Reason: g varies due to Earth's shape and rotation.

43. Correct answer: (A) r_1/r_2

Reason: $g \propto R$ when densities equal.

44. Correct answer: (B)

Reason: 9.8 m/s^2 is the average value of g.

45. Correct answer: (B) $g - r\omega^2$

Reason: At equator centrifugal force reduces g.

46. Correct answer: (D) $m\omega^2 r$

Reason: Increase in weight equals centrifugal term difference.

47. Correct answer: (B) $\sqrt{(g/r)}$

Reason: Weightless condition: $g = \omega^2 r \rightarrow \omega = \sqrt{(g/r)}$.

INTEGER TYPE

48. Correct answer: 3

Given same density and radius = 3R.
Since $g \propto R$ (for same density),
$g'/g = 3$.

49. Correct answer: 10 m/s

Given: u = 0, g = 10 m/s², t = 1 s
v = u + gt = 0 + (10 × 1) = 10 m/s

50. Correct answer: 2 s

Given: v = 20 m/s, g = 10 m/s²
$v = gt \Rightarrow t = v/g = 20/10 = 2$ s

Previous Year Questions

1. Correct answer: (A)

For configuration P,

$$F_P = \frac{GMm}{a^2} + \frac{GMm}{(2a)^2} = \frac{5}{4}\frac{GMm}{a^2}$$

For configuration Q,

$$F_Q = \frac{GMm}{a^2} - \frac{GMm}{a^2} = 0$$

For configuration R,

$$F_R = \sqrt{F_1^2 + F_2^2}$$

$$= \sqrt{\left(\frac{GMm}{a^2}\right)^2 + \left(\frac{GMm}{a^2}\right)^2} = \frac{GMm\sqrt{2}}{a^2}$$

Thus, greatest to least magnitude of gravitational force on M are

$$R > P > Q.$$

2. Correct answer (D) :

By Kepler's second law of planetary motion, areal velocity of planet remains constant.

Thus,

$$\frac{a_1}{3} = \frac{a_2}{4} = \frac{a_3}{12}$$

$4a_1 = 3a_2 = a_3$

3. Correct answer (D):

I. It is true.

II. It is false, as they appear to move from East to West.

III. It is false, as INSAT was not first Indian satellite.

IV. It is false, there is no any such fix time interval for meteor shower

4. Correct answer: (B) : As, $F \propto m_1 m_2 / r^2$

(i) inversely, (ii) square

(iii) $F = mg = 1 \times 9.8$ N = 9.8 N

(iv) $F = G\, m_1 m_2 / r^2$

$F' = G \cdot m_1 m_2 / (0.9r)^2$

percentage change in force = $\frac{F'-F}{F} \times 100$

$$= \left(\frac{Gm_1m_2r^2}{0.81r^2 \times Gm_1m_2} - 1\right)100 = 23.5\%$$

5. Correct answer: (A)

Gravitational force between two masses is independent of medium in which they are placed.

6. Correct answer: (B)

Acceleration due to gravity at the surface of earth,

$$g = \frac{GM_e}{R_e^2}$$

If universal gravitational constant (G) becomes ten times larger in magnitude, rain drops will fall faster, walking on ground will become more difficult, time period of simple pendulum on the earth will decrease. All of this is because acceleration due to gravity on the surface of earth will increase.

7. Correct answer (D):

Gravitational force between two particles,

$$F = \frac{Gm_1m_2}{r^2}$$

Where m_1 and m_2 are the mass of particles and r is the distance between them.

If a_1 is the acceleration of m_1, then

$$m_1 a_1 = \frac{G m_1 m_2}{r^2}$$
$$a_1 = \frac{G m_2}{r^2}$$

Similarly, acceleration of m_2,

$$a_2 = \frac{G m_1}{r^2}$$

So,

$$\frac{a_1}{a_2} = \frac{m_2}{m_1}$$

Therefore acceleration of both the particles are unequal and acceleration of particle of mass m_1 is proportional to m_2.

8. Correct answer: (A)

Mass of ball, m = 100 g = 0.1 kg
For first ball, u_1 = 49 m/s
For second ball, s = 98 m

If they collide after time t above height x from ground.

$x = 49t - \frac{1}{2} \times 9.8t^2$ (for first ball)… (i)

$98 - x = \frac{1}{2} \times 9.8t^2$ (for second ball)… (ii)

Adding equations (i) and (ii), we get

$$98 = 49t$$
$$\Rightarrow t = 2\ s$$

After t = 2 s,

For first ball,

$$v_1 = 49 - 9.8 \times 2$$
$$\Rightarrow v_1 = 49 - 19.6$$
$$= 29.4\ m\ s^{-1} \text{(upwards)}$$

For second ball,

$$v_2 = 9.8 \times 2 = 19.6\ m\ s^{-1} \text{(downwards)}$$

If the combined mass moves with velocity $v\ m\ s^{-1}$ then, from the law of conservation of momentum

$$(m_1 + m_2)v = m_1 v_1 + m_2 v_2$$
$$0.2v = 0.1 \times 29.4 + 0.1(-19.6)$$

v = 0.98 / 0.2 = 4.9 m s^{-1}

v = 4.9 m s^{-1}

4. FLUID MECHANIC

Solutions

Worksheet – 1

1. **Correct answer: (A) Gas molecules interact weakly and gases are highly compressible**
   Explanation: Gases have large intermolecular spaces, so volume decreases easily under pressure.
2. **Correct answer: (B) A force acts upward called buoyant force (upthrust)**
   Explanation: Higher pressure at the bottom creates a net upward force (upthrust).
3. **Correct answer: (A) 2.45×10^5 Pa**
   Explanation: Gauge pressure $P = \rho gh = 1000 \times 9.8 \times 25 = 2.45 \times 10^5$Pa.
4. **Correct answer: (C) $1.993\times10^+$ Pa**
   Explanation: Absolute pressure $P = P_{atm} + \rho gh = 1.013 \times 10^5 + 1000 \times 9.8 \times 10 = 1.993 \times 10^5$Pa.
5. **Correct answer: (A) 1000 kg/m³**
   Explanation: $\rho = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2} = \frac{900\times2+1200\times1}{3} = 1000.$
6. **Correct answer: (A) 980 N**
   Explanation: Thrust $F = \rho gh \;\; A = 1000 \times 9.8 \times 5 \times 0.02 = 980$N.
7. **Correct answer: (B) Hydrostatics**
   Explanation: Fluids at rest are studied in hydrostatics (fluid statics).
8. **Correct answer: (A) 60 N**
   Explanation: Buoyant force $= (\boldsymbol{P_{bottom} - P_{top}})\boldsymbol{A} = (\mathbf{6500 - 5000}) \times \mathbf{0.04 = 60N.}$
9. **Correct answer: (A) 800 kg/m³**
   Explanation: Density $\rho = \frac{m}{V} = \frac{4.8}{0.006} =$ 800kg/m³.
10. **Correct answer: (B) 2.189×10^5 Pa**
    Explanation: $P_{abs} = P_{atm} + \rho gh = 1.013 \times 10^5 + 1000 \times 9.8 \times 12 = 2.189 \times 10^5$Pa.

Worksheet-2

1. **Correct answer: (B) 5880 Pa**
   Explanation: The pressure difference depends on the depth difference.
   $\Delta P = \rho g \Delta h$
   $= 1000 \times 9.8 \times 0.6 = 5880$ Pa
2. **Correct answer: (B) There is vacuum above the mercury column**
   Explanation: In a mercury barometer, the tube is sealed at the top. When mercury falls to balance atmospheric pressure, a space is left above the mercury column. This space contains no air and is called Torricellian vacuum, so the pressure there is nearly zero.
3. **Correct answer: (B) torr**
   Explanation: The unit "mm Hg" is known as **torr**, named after Evangelista Torricelli.
   1 mm Hg ≈ 1 torr
4. **Correct answer: (B) 120 N**
   Explanation: According to Pascal's law, pressure remains the same in a hydraulic system. $\frac{F_1}{A_1} = \frac{F_2}{A_2}$
   $F_1 = 12000 \times \frac{0.003}{0.30} = 120$ N
5. **Correct answer: (B) Pressure is transmitted equally in all directions**
   Explanation: This is a direct application of Pascal's law, which states that pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions.
6. **Correct answer: (C) 0.76 m**
   Explanation: At atmospheric pressure,
   $$h = \frac{P}{\rho g}$$
   $$h = \frac{1.013 \times 10^5}{13600 \times 9.8} \approx 0.76 \text{ m}$$
   (= 76 cm of mercury)

7. **Correct answer: (D) 2000 N**
Explanation: Using the hydraulic lift principle,
$$F_2 = F_1 \times \frac{A_2}{A_1}$$
$$F_2 = 40 \times \frac{0.10}{0.002} = 2000 \text{ N}$$
(Minor note: Unit of $F_1$is implied as newton.)

8. **Correct answer:(C) increases by the same amount**
Explanation: Absolute pressure is given by
$$P_{abs} = P_{atm} + \rho gh$$
At a fixed depth, $\rho gh$is constant. Therefore, any increase in atmospheric pressure increases absolute pressure by the same amount.

9. **Correct answer: (C) 6 m**
Explanation: Gauge pressure:
$$P_g = P_{abs} - P_{atm}$$
$$= 1.60 \times 10^5 - 1.013 \times 10^5$$
$$= 0.587 \times 10^5 \text{ Pa}$$
$$h = \frac{P_g}{\rho g} = \frac{58700}{1000 \times 9.8} \approx 5.99 \text{ m}$$

10. **Correct answer: (B) 9.60×10^4Pa**
Explanation: $P = \rho gh$
Convert height: 72 cm = 0.72 m
$P = 13600 \times 9.8 \times 0.72 \approx 9.60 \times 10^4$ Pa

Worksheet-3

1. **Correct answer: (B) 98 N**
Explanation: Buoyant force $F_B = \rho Vg = \rho(A \times h)g = 1000(0.020 \times 0.50) \times 9.8 = 98\,N$. Correct value = 98 N.

2. **Correct answer: (B) 5: 4**
Explanation: $F_B = \rho Vg \Rightarrow \frac{F_{water}}{F_{ker}} = \frac{\rho_w}{\rho_k} = \frac{1000}{800} = \frac{5}{4}$
3 :

3. **Correct answer: (A)**

4. **Correct answer: (C) Weight of the fluid displaced by the immersed part**
Explanation: Archimedes' principle: $F_B =$ weight of displaced fluid $= \rho Vg$

5. **Correct answer: (B) Air exerts an upthrust on the object**
Explanation: Apparent weight $W_{app} = W_{true} - F_B$, and in air $F_B = \rho_{air} Vg \neq 0$

6. **Correct answer: (C) 5**
Explanation: Relative density $= \frac{W_{air}}{W_{air} - W_{water}} = \frac{50}{50-40} = \frac{50}{10} = 5$

7. **Correct answer: (B) 0.6**
Explanation: $F_B$in water $= 80 - 60 = 20N$. In unknown liquid $= 80 - 68 = 12N$.
RD of liquid $= \frac{12}{20} = 0.6$

8. **Correct answer: 8 (A)**
Loss of weight in water
$$W_1 - W_2 = 50 - 44 = 6 \text{ N}$$
Buoyant force $= \rho_w g V_{\text{outer}}$
$$6 = 1000 \times 10 \times V_{\text{outer}} \Rightarrow V_{\text{outer}} = 6.0 \times 10^{-4} \text{ m}^3$$
Volume of metal material
$$V_{\text{metal}} = \frac{M}{\sigma} = \frac{5}{10^4} = 5.0 \times 10^{-4} \text{ m}^3$$
Volume of cavity
$$V_{\text{cavity}} = V_{\text{outer}} - V_{\text{metal}}$$
$$= (6.0 - 5.0) \times 10^{-4}$$
$$= 1.0 \times 10^{-4} \text{ m}^3$$

9. **Correct answer: (C) Buoyancy**
Explanation: Upward force on immersed body is buoyant force due to buoyancy: $F_B = \rho Vg$

10. **Correct answer: (C) Weight = upthrust produced by displaced liquid**
Explanation: Floating condition: $W = F_B = \rho V_{disp} g$

Worksheet-4

1. **Correct answer: (B) an object floats at different levels in different liquids depending on liquid density**
Explanation: Float level depends on density because $F_B = \rho Vg$and floating needs $W = F_B$

2. **Correct answer: (B) The milk contains added water, reducing density**
Explanation: Lower density ⇒ less upthrust per volume $F_B = \rho Vg \Rightarrow$ lactometer sinks more toward water mark

3. **Correct answer: (B) 0.25**
Explanation: Fraction submerged $= \frac{\rho_{body}}{\rho_{liq}} = \frac{750}{1000} = 0.75$. Outside fraction $= 1 - 0.75 = 0.25$

4. **Correct answer: (A) A denser liquid produces more buoyant force for the same displaced volume**

   Explanation: $F_B = \rho V g$. Higher $\rho \Rightarrow$ higher $F_B \Rightarrow$ less volume needed to balance weight $\Rightarrow$ floats higher

5. **Correct answer: (B) 823.5 kg/m³**

   Explanation: Equal masses m:
   Total volume $= \frac{m}{1000} + \frac{m}{700}$.
   $\rho_{mix} = \frac{2m}{m(\frac{1}{1000}+\frac{1}{700})} = \frac{2}{0.001+0.0014286} = 823.5$

6. **Correct answer: (C) 0.010 m³**

   Explanation: Submerged fraction $= \frac{\rho_{block}}{\rho_{water}} = \frac{500}{1000} = 0.5$.
   Submerged volume $= 0.5 \times 0.020 = 0.010$.
   Outside volume $= 0.020 - 0.010 = 0.010$

7. **Correct answer: (C) 6 N**

   Explanation: Extra displaced volume $= A\Delta h = 0.06 \times 0.01 = 6 \times 10^{-4}\ m^3$.
   Extra upthrust $= \rho g V = 1000 \times 10 \times 6 \times 10^{-4} = 6\ N$

8. **Correct answer: (B) A will float lower than B**

   Explanation: Higher density $\Rightarrow$ more weight $\Rightarrow$ must displace more volume to satisfy $W = F_B = \rho V g \Rightarrow$ sinks more

9. **Correct answer: (C) 50%**

   Explanation: Fraction submerged $= \frac{\rho_{block}}{\rho_{liq}} = \frac{400}{800} = 0.5 = 50\%$

10. **Correct answer: (B) Milk density has decreased because of added water**

    Explanation: Lower $\rho \Rightarrow$ smaller $F_B = \rho V g$ for same immersion $\Rightarrow$ lactometer sinks closer to water mark

Worksheet-5

1. **Correct answer:(B) along the tangent to the streamline at that point**
   Explanation: By definition, a streamline is a curve drawn in such a way that the tangent at any point on it gives the direction of the velocity of the fluid particle at that point.

2. **Correct answer: (B) Higher velocity where streamlines are crowded**
   Explanation: According to the equation of continuity,
   $$Av = \text{constant}$$
   Where streamlines are closer together, the effective cross-sectional area of flow is smaller, so the velocity of the fluid is higher.

3. **Correct answer: (C) 6 m/s**
   Explanation: Using the equation of continuity,
   $$A_1 v_1 = A_2 v_2$$
   $v_2 = \frac{A_1 v_1}{A_2} = \frac{0.020 \times 3}{0.010} = 6\ \text{m/s}$

4. **Correct answer: (D) 2.3×10^5 Pa**
   Explanation: First, find v_2 using continuity:
   $A_1 v_1 = A_2 v_2$
   $0.03 \times 2 = 0.01 \times v_2 \Rightarrow v_2 = 6\ \text{m/s}$
   Using Bernoulli's equation for a horizontal pipe,
   $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$
   $P_2 = 2.5 \times 10^5 - \frac{1}{2}(1000)(6^2 - 2^2)$
   $P_2 = 250000 - 16000 = 234000\ \text{Pa}$
   $P_2 \approx 2.3 \times 10^5\ \text{Pa}$

5. **Correct answer: (C)**

6. **Correct answer: (C) higher pressure below the wing pushing it up**
   Explanation: According to Bernoulli's principle, the speed of air above the wing is higher, so the pressure there is lower. Below the wing, air moves slower and hence has higher pressure. The pressure difference produces an upward lift force on the wing.

7. **Correct answer: (C) 3 m/s**
   Explanation:
   Using conservation of mass,
   $A_{\text{main}} v_{\text{main}} = A_1 v_1 + A_2 v_2$
   $(0.03 \times 2) = (0.01 \times 3) + (0.01 \times v_2)$
   $0.06 = 0.03 + 0.01 v_2$
   $v_2 = 3\ \text{m/}$

8. **Correct answer: (C) $P + \rho g h + \frac{1}{2}\rho v^2 =$ constant**
   Explanation: This is the mathematical statement of Bernoulli's theorem. It

represents conservation of mechanical energy per unit volume for steady flow of an ideal fluid.

9. **Correct answer: (C) 3 m/s**
Explanation:
According to Torricelli's law, $v = \sqrt{2gh}$
$v = \sqrt{2 \times 10 \times 0.45} = \sqrt{9} = 3$ m/s

10. **Correct answer: (B) air is somewhat viscous, causing friction**
Explanation: The Magnus effect occurs due to the viscosity of air. Viscosity allows the spinning ball to drag air around it, producing different air speeds on opposite sides. This creates a pressure difference, resulting in a sideways force.

COMPETETIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct answer: (A) Increases by 2%**
Explanation: The height of a barometric column is given by $h = \frac{P_{atm}}{\rho g}$
Thus, $h \propto \frac{1}{g}$. If the value of acceleration due to gravity decreases by 2%, the height of the mercury column increases by approximately 2% to maintain the same atmospheric pressure.

2. **Correct answer: (D) The pressure on the bottom of tank A and tank B is the same**
Explanation: According to the hydrostatic paradox, the pressure at a given depth in a liquid depends only on the height of the liquid column, density of the liquid, and acceleration due to gravity. $P = \rho g h$
It does not depend on the shape or cross-sectional area of the container. Hence, the pressures at the bottom of both tanks are equal.

3. **Correct answer: (A) 76 cm of Hg**
Explanation: Standard atmospheric pressure at sea level is defined as one atmosphere (1 atm), which is equal to the pressure exerted by a mercury column of height 760 mm or 76 cm.

4. **Correct answer: (C) $P_1 < P_2$**
Explanation: Atmospheric pressure increases with depth because the weight of the air column above a point increases. Since a mine is located below the Earth's surface, the pressure inside the mine (P_2) is greater than the pressure at the surface (P_1).

5. **Correct answer: (B) The atmospheric pressure decreases**
Explanation: At high altitude, external pressure P_{atm} decreases, so blood pressure inside capillaries can exceed outside pressure causing nose bleed

6. **Correct answer: (D) 14 m/s**
Explanation: Torricelli's law $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 10} = \sqrt{196} = 14$ m/s

7. **Correct answer: (C) Remains unchanged**
Explanation: When the man drinks water, the weight of the water is transferred from the pond to the boat (via the man). A floating body displaces a weight of fluid equal to its own weight. Therefore, the volume of water removed from the pond is exactly replaced by the increase in the volume of water displaced by the boat.

8. **Correct answer: (B)**
Explanation: For floating equilibrium:
$F_B = W = mg$

9. **Correct answer: (C) The balance A will read less than 2 kg, B will read more than 5 kg**
Explanation:
Balance A: The immersed block experiences an upward buoyant force (U), so the reading of balance A becomes
$1 \text{ kg} - \frac{U}{g}$
which is less than 1 kg, and therefore less than 2 kg.
Balance B: By Newton's third law, the block exerts an equal downward force on the liquid. Hence, the reading of balance B becomes $5 \text{ kg} + \frac{U}{g}$
which is greater than 5 kg.

10. **Correct answer: (C) $(H - h)\rho g$**
Explanation: Gauge pressure is determined by the depth below the free surface of the liquid. Depth below the free surface

$=$ Total height (H)
$-$ height from bottom (h)
$= H - h$
Therefore, Gauge pressure = Depth × $\rho \times g = (H - h)\rho g$

11. Correct answer: (A) 4.625×10^3 kg-wt/m^2
Explanation: Total pressure at the bottom is the sum of pressures due to both liquids.
$P = P_{\text{water}} + P_{\text{oil}}$
$P = h_w\rho_w + h_o\rho_o (\text{in kg-wt/m}^2)$
$P = (2.5 \times 1000) + (2.5 \times 850)$
$P = 2500 + 2125 = 4625 \text{ kg-wt/m}^2$
$= 4.625 \times 10^3 \text{ kg-wt/m}^2$

12. Correct answer: (A) $6 \times 1000 \times 10 \times 80$ N
Explanation: Thrust (force) on the bottom is given by Thrust = Pressure × Area
Pressure, $P = \rho gh = 1000 \times 10 \times 6$
Area, A = Length × Breadth $= 10 \times 8 = 80$ m^2
Therefore, Thrust $= (1000 \times 10 \times 6) \times 80$

13. Correct answer: (C) 1250 kgf
Explanation: According to Pascal's principle for a hydraulic machine, $\frac{F_1}{A_1} = \frac{F_2}{A_2}$
Since area is proportional to the square of diameter, $\frac{F_1}{d_1^2} = \frac{F_2}{d_2^2}$

$$F_2 = F_1\left(\frac{d_2}{d_1}\right)^2$$
$$F_2 = 50 \times \left(\frac{25}{5}\right)^2$$
$$F_2 = 50 \times 25 = 1250 \text{ kgf}$$

14. Correct answer: (B) 20 m
Explanation: Let atmospheric pressure be P_0(equivalent to 10 m water column) and total depth be h.
Pressure at half depth, $P_{1/2} = P_0 + \frac{h}{2}$
Pressure at bottom, $P_{\text{bottom}} = P_0 + h$
Given, $P_0 + \frac{h}{2} = \frac{2}{3}(P_0 + h)$
Solving, $3P_0 + 1.5h = 2P_0 + 2h$
$P_0 = 0.5h \Rightarrow h = 2P_0$
Since $P_0 \approx 10$ m,
$h = 20$ m

15. Correct answer: (A) $P_1 < P_2$
Explanation: Hydrostatic pressure is given by $P = \rho gh$
Since the density of seawater is greater than that of river water, the pressure at the same depth in seawater is higher.
Hence, $P_2 > P_1$

16. Correct answer: (D) 100%
Explanation: The wood floats with 50% of its volume immersed in water.
Therefore, the density of the wood is $\rho_{\text{wood}} = 0.5 \times \rho_{\text{water}} = 0.5$ g/cm^3
When placed in a liquid of density 0.5 g/cm^3, the density of the wood equals the density of the liquid. Hence, the wood is in neutral equilibrium and remains fully submerged. Therefore, 100% of its volume is immersed.

17. Correct answer: (B) 11%
Explanation: By the law of floatation, the fraction of volume submerged is
$\frac{V_{\text{sub}}}{V_{\text{total}}} = \frac{\rho_{\text{ice}}}{\rho_{\text{seawater}}}$; $\frac{0.92}{1.03} \approx 0.893$
The fraction of ice above water is $1 - 0.893 = 0.107 \approx 10.7\% \approx 11\%$

18. Correct answer: (B) 15 cm^3
Explanation: Loss of weight in water (upthrust) = 40 g − 20 g = 20 g
Since density of water is 1 g/cm^3,
Total volume of sphere = 20 cm^3
Volume of metal, $= \frac{\text{Mass}}{\text{Density}} = \frac{40}{8} = 5$ cm^3
Volume of cavity $= 20 - 5 = 15$ cm^3

19. Correct answer: (A) $AV = A_1V_1 + A_2V_2$
Explanation: According to the equation of continuity for an incompressible fluid, the volume flow rate entering the pipe is equal to the sum of the flow rates leaving through the branches.
$Q_{\text{in}} = Q_{\text{out}}$
$AV = A_1V_1 + A_2V_2$

20. Correct answer: (D) $A_1v_1 = A_2v_2$
Explanation: For an ideal liquid flowing through a tube of varying cross-sectional area, the flow rate remains constant. Hence, the product of cross-sectional area and velocity is constant. $A_1v_1 = A_2v_2$

21. Correct answer: (B) 10 ms^{-1}
Explanation: $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 5} = 10$

22. Correct answer: (A) $\frac{4V}{\pi d^2}$
Explanation: $V = Av$, $A = \frac{\pi d^2}{4} \Rightarrow v = \frac{V}{A} = \frac{4V}{\pi d^2}$

23. Correct answer: (B) 13.75 Pa
Explanation: $A_2 = 1.2A_1 \Rightarrow v_2 = \frac{v_1}{1.2} = 0.25$.
$\Delta P = \frac{1}{2}\rho(v_1^2 - v_2^2) = \frac{1}{2}(1000)(0.09 - 0.0625) = 13.75$Pa

24. Correct answer: (C) 1 : 4
Explanation: Extra displaced fraction = $\frac{5}{6} - \frac{2}{3} = \frac{1}{6}$.
$$\frac{m_{man}}{m_{wood}} = \frac{1/6}{2/3} = \frac{1}{4} \Rightarrow 1:4$$

25. Correct answer: (D)

26. Correct answer: (C) $\sqrt{400}$ m s^{-1}
Explanation: The velocity of efflux depends on the pressure difference between the inside and outside of the hole.
Pressure inside the tank = 3 atm
Atmospheric pressure outside = 1 atm
$\Delta P = 3 - 1 = 2$ atm $= 2 \times 10^5$ Pa
Using Bernoulli's equation,
$\frac{1}{2}\rho v^2 = \Delta P$
$v = \sqrt{\frac{2\Delta P}{\rho}}$ $v = \sqrt{\frac{2 \times 200000}{1000}} = \sqrt{400}$
$= 20$ m/s

27. Correct answer: (D) 4 m s^{-1}
Explanation: According to the equation of continuity, $A_1 v_1 = A_2 v_2$
Since area $A \propto r^2$, doubling the radius makes the area four times. $A_2 = 4A_1$
$$v_2 = \frac{v_1}{4} = \frac{16}{4} = 4 \text{ m/s}$$

28. Correct answer: (D) Density of liquid
Explanation: The volume of liquid emerging per second (flow rate) is $Q = A \times v$
By Torricelli's law, $v = \sqrt{2gh}$; $Q = A\sqrt{2gh}$
This expression depends on area A, gravity g, and height h, but does not involve the density of the liquid.

29. Correct answer: (C) 8×10^4 N
Explanation: Lift force is given by $F = \Delta P \times A$
$\Delta P = (9.1 - 9.0) \times 10^4 = 0.1 \times 10^4$
$= 1000$ Pa
$F = 1000 \times 80 = 80{,}000$ N
$F = 8 \times 10^4$ N

30. Correct answer: (C) 10^{-5} m^3
Explanation: Velocity of efflux, $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 5} = \sqrt{100} = 10$ m/s
Area of hole, $A = 1$ mm^2 $= 10^{-6}$ m^2
Volume of water flowing per second, $Q = A \times v = 10^{-6} \times 10 = 10^{-5}$ m^3/s

31. Correct answer: B and D
Explanation (short & clear):
In equilibrium, the pressure transmitted through the fluid must be the same at the same level (Pascal's law), so
$P_1 = P_2 \Rightarrow$ (Option D correct)
Pressure $P = \frac{F}{A} = \frac{mg}{a}$for the small piston and $\frac{Mg}{10a}$for the large piston.
Equating pressures:
$$\frac{mg}{a} = \frac{Mg}{10a} \Rightarrow M = 10m$$

32. Correct answer: All four options (A), (B), (C), and (D) are correct.
Explanation: A) Pressure at the base
Hydrostatic pressure depends only on the vertical depth of the liquid column.
Total height of liquid $H = h + h = 2h$.
Neglecting atmospheric pressure, $P = \rho g H = \rho g(2h) = 2h\rho g$
∴ Statement A is correct.
B) Force exerted on the base
Force exerted by a liquid on a surface equals pressure at that surface multiplied by area.
ressure at base $= 2h\rho g$
Area of base $= A_2$
$F_{\text{base}} = P \times A_2 = 2h\rho g A_2$
∴ Statement B is correct.
C) Weight of the liquid
Weight = Volume × ρ × g.
Volume of upper part $= A_1 h$
Volume of lower part $= A_2 h$
$W = (A_1 h + A_2 h)\rho g = h\rho g(A_1 + A_2)$
Since $A_1 < A_2$,

$(A_1 + A_2) < 2A_2 \Rightarrow W < 2h\rho gA_2$

$\therefore$ Statement C is correct.

D) Force at level x

Depth of liquid at level $x = h$.

Pressure at that level,

$$P_x = h\rho g$$

This pressure acts upward on the horizontal shoulder area $(A_2 - A_1)$.

Downward force on the liquid (by the walls),

$$F = P_x(A_2 - A_1) = h\rho g(A_2 - A_1)$$

$\therefore$ Statement D is correct.

33. Correct answer: Fluid in an Artificial Satellite

Correct Options: A and C

Explanation:

A) The points at different depths have the same pressure

In an artificial satellite, the system is in a state of free fall, so the effective gravity

$$g_{\text{eff}} = 0$$

Hydrostatic pressure difference is given by

$$\Delta P = \rho g_{\text{eff}} h$$

Since $g_{\text{eff}} = 0$,

$$\Delta P = 0$$

Hence, pressure is the same at all depths.

$\therefore$ Statement A is correct.

C) Acceleration due to gravity is not zero

Although the satellite experiences weightlessness, Earth's gravitational force is still acting on it. This gravitational pull provides the centripetal force required for orbital motion.

Thus, gravity is not absent; only its effect is cancelled due to free fall.

$\therefore$ Statement C is correct.

34. Correct answer: Venturi Tube Flow

A, C, and D

Explanation: A) The energy per unit mass is the same

For steady flow through a horizontal pipe, Bernoulli's principle applies:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Since the pipe is horizontal, potential energy remains constant, so total energy per unit mass remains the same.

$\therefore$ Statement A is correct.

C) $V_2^2 - V_1^2 = 2gh$

From Bernoulli's equation: $P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$

Given pressure difference from vertical tubes: $P_1 - P_2 = \rho gh$

Substituting, $\frac{1}{2}\rho(V_2^2 - V_1^2) = \rho gh \Rightarrow V_2^2 - V_1^2 = 2gh$

$\therefore$ Statement C is correct.

D) The volume flowing in unit time is A_1V_1

According to the equation of continuity, $Q = Av =$ constant

Thus, volume flow rate equals A_1V_1.

$\therefore$ Statement D is correct.

35. Correct answer: Rate of Leaking B

Explanation: Velocity of efflux (Torricelli's law): $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 5} = \sqrt{100} = 10$ m/s

Area of the hole: $1\ \text{cm}^2 = 10^{-4}\ \text{m}^2$

Rate of flow: $Q = A \times v = 10^{-4} \times 10 = 10^{-3}\ \text{m}^3/\text{s}$

$\therefore$ Correct option is B.

ONE OR MORE THAN ONE CORRECT

36. Correct answer: (C) A is true but R is false

Explanation: Hydraulic lift works by Pascal's law: $P = \frac{F}{A}$transmitted equally $\Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$.

R is wrong because pressure is transmitted equally in all directions, not "only one direction".

37. Correct answer: (A) Both A and R are true and R is the correct explanation of A

Explanation: $P_{inside} < P_{atm} \Rightarrow$ net force $F = (P_{atm} - P_{inside})A$presses cup to wall

38. Correct answer: (A) Both A and R are true and R is the correct explanation of A

Explanation: Higher density $\Rightarrow$ more upthrust per volume $F_B = \rho Vg \Rightarrow$ lactometer floats higher $\Rightarrow$ higher reading

39. Correct answer: (D) A is false but R is true

Explanation: Ship floats because its average density < water (so it does NOT sink).

R is true: floating when $F_B = \rho_w g V_{disp} = W$.

Level-4

40. Correct answer:

a) A body will sink only when

→ This happens when density of body is greater than density of liquid

Match: R) $\rho_B > \rho_L$

b) Apparent weight of the floating body

→ For a floating body, buoyant force equals weight, so apparent weight is zero

Match: S) Zero

c) The body will float fully submerged in neutral equilibrium

→ Neutral equilibrium occurs when densities are equal

Match: T) $\rho_B = \rho_L$

d) The body will move upwards and in equilibrium will float partially immersed

→ This happens when buoyant force is greater than weight

Match: Q) $W <$ thrust

41. Correct answer:

a) Dynamic lift → R) $\frac{1}{2}\rho(v_2^2 - v_1^2)A$

Explanation: Dynamic lift arises due to the difference in velocities of fluid on two sides of a body. The resulting pressure difference produces a lift force given by $\frac{1}{2}\rho(v_2^2 - v_1^2)A$.

b) Equation of continuity → P) $av =$ constant

Explanation: The equation of continuity states that for incompressible flow, the product of cross-sectional area and fluid velocity remains constant, ensuring conservation of mass.

c) Fluid flux rate in laminar flow → T) Rate of flow of mass is constant

Explanation: In steady laminar flow, the mass of fluid passing through any cross-section per unit time remains the same.

d) Bernoulli's theorem → Q) $P + \frac{1}{2}\rho v^2 + \rho g h =$ constant

Explanation: Bernoulli's theorem relates pressure, kinetic energy per unit volume, and potential energy per unit volume of a fluid, stating that their sum remains constant along a streamline.

Passage

Given Data

- Area at **A** = 12 $\text{cm}^2 = 12 \times 10^{-4}$ m^2
- Area at **B** = 5.6 cm^2
- Area at **C** = 6 $\text{cm}^2 = 6 \times 10^{-4}$ m^2
- Speed at **C**, $v_C = 8\ \text{m s}^{-1}$
- Atmospheric pressure $P_{\text{atm}} = 1.01 \times 10^5$ Pa
- Density of mercury $\rho = 13.6 \times 10^3\ \text{kg m}^{-3}$
- $g = 10\ \text{m s}^{-2}$

42. Correct answer: Speed of mercury at point A

Using **equation of continuity**:

$A_A v_A = A_C v_C$

$$v_A = \frac{A_C}{A_A} v_C = \frac{6}{12} \times 8 = 4\ \text{m s}^{-1}$$

(B) 4 m/s

43. Correct answer: Pressure at point A

Apply **Bernoulli's theorem** between points A and C (same horizontal level):

$$P_A + \frac{1}{2}\rho v_A^2 = P_C + \frac{1}{2}\rho v_C^2$$

Since point C is open to atmosphere:

$P_C = 1.01 \times 10^5$ Pa

$P_A = P_C + \frac{1}{2}\rho(v_C^2 - v_A^2)$

$P_A = 1.01 \times 10^5 + \frac{1}{2}(13.6 \times 10^3)(64 - 16)$

$P_A = 1.01 \times 10^5 + 3.26 \times 10^5$

$P_A \approx 4.27 \times 10^5$ Pa

(C) 4.27×10^5 Pa

44. Correct answer: Height of mercury column in manometer

Pressure difference: $\Delta P = P_A - P_{\text{atm}} = 4.27 \times 10^5 - 1.01 \times 10^5 = 3.26 \times 10^5$ Pa

Using manometer relation: $\Delta P = \rho g h$

$$h = \frac{3.26 \times 10^5}{13.6 \times 10^3 \times 10} = 2.39 \times 10^{-1}\ \text{m} \approx 0.263\ \text{m}$$

$h = 263$ mm

(C) 263 mm

Passage-2

45. Correct answer:

Question: Water flows through a pipe of radius

$$r = \left(\frac{2}{\sqrt{\pi}}\right) \text{ cm}$$

with a speed of 10 m/s. Find the mass of water flowing per second through the pipe.

Given:

Density of water, $\rho = 1000$ kg/m^3

Radius, $r = \frac{2}{\sqrt{\pi}}$ cm

Velocity, $v = 10$ m/s

Step 1: Convert radius into metres

$r = \frac{2}{\sqrt{\pi}} \times 10^{-2}$ m

Step 2: Area of cross-section

$$A = \pi r^2$$
$$A = \pi\left(\frac{2}{\sqrt{\pi}} \times 10^{-2}\right)^2$$
$$A = \pi\left(\frac{4}{\pi} \times 10^{-4}\right)$$
$$A = 4 \times 10^{-4} \text{ m}^2$$

Step 3: Mass flow rate

$$\frac{dm}{dt} = \rho A v$$
$$\frac{dm}{dt} = 1000 \times 4 \times 10^{-4} \times 10$$
$$\frac{dm}{dt} = 4 \text{ kg/s}$$

(B) 4 kg/s

46. Correct answer:

Question: Water flows at a speed of 10 m/sthrough a pipe of cross-sectional area 0.04 m^2. If the area is reduced to half, find the new rate of flow (velocity).

Given: Initial velocity,

$$v_1 = 10 \text{ m/s}$$

Initial area, $A_1 = 0.04$ m^2

Final area, $A_2 = \frac{A_1}{2} = 0.02$ m^2

Step 1: Apply equation of continuity

$A_1 v_1 = A_2 v_2$

Step 2: Substitute values

$$0.04 \times 10 = 0.02 \times v_2$$
$$0.4 = 0.02 v_2$$
$$v_2 = 20 \text{ m/s}$$

(D) 20 m/s

47. Correct answer:

Question: In a tube of variable cross-section, liquid enters with a velocity of 5 m/sfrom a wider end and leaves from a narrow end whose radius is $\frac{1}{4}$th of the wider end. Find the exit velocity.

Given: $v_1 = 5$ m/s ; $r_2 = \frac{1}{4} r_1$

Step 1: Relate areas to radii

$A_1 = \pi r_1^2$

$$A_2 = \pi r_2^2 = \pi\left(\frac{r_1}{4}\right)^2$$
$$A_2 = \frac{A_1}{16}$$

Step 2: Apply equation of continuity $A_1 v_1 = A_2 v_2$

$$A_1 \times 5 = \frac{A_1}{16} \times v_2$$

Step 3: Solve

$5 = \frac{v_2}{16}$

$$v_2 = 80 \text{ m/s}$$

Answer: (C) 80 m/s

48. Correct answer: 2

Explanation: Using the Work–Energy Theorem:

Work by gravity + Work by buoyancy $= \Delta KE = 0$

$mg(h + d) - F_B d = 0$

$V\rho g(h + d) = V\sigma g d$

Substituting values:

$400(9 + d) = 1000d$

$3600 = 600d \Rightarrow d = 6$ cm

Given depth $= n \times 3$:

$6 = 3n \Rightarrow n = 2$

49. Correct answer: 2

Explanation: According to Torricelli's law,

$v = \sqrt{2gh}$

$v = \sqrt{2 \times 10 \times 1} = \sqrt{20} = 2\sqrt{5}$ m/s

Comparing with the form $_ \times \sqrt{5}$, the value is **2**.

50. Correct answer: 1

Explanation: The time taken by the liquid jet to strike the base is the time of flight of a horizontally projected body.

Height of hole from base: $y =$ Total height $-$ Depth $= 10 - 5 = 5$ m

Using: $y = \frac{1}{2} g t^2$

$$5 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t^2 = 1 \Rightarrow t = 1 \text{ s}$$

Previous Year Questions

1. Correct answer:

air pressure at the base of the mountain is given by P=hdg

Thus, P= (96.4-20) x 10^{-2} x 13600 x 10

=1.04 x 10^{5} Nm^{-2}

2. Correct answer: (D)

l decreases as block moves up due to reduction in weight after coin fall into water. *h* also decreases because when coin is in water it displaces water equal to its volume, whereas when it is on the block it displaces water equal to its weight.

3. Correct answer:(C)

If $\frac{1}{4}^{\text{th}}$ volume is outside, so $\frac{3}{4}^{\text{th}}$ volume is inside.

For water, $v\sigma g = \frac{3}{4}\, v\rho g \ldots (i)$

where, σ is density of body, ρ is density of water.

For liquid, $v\sigma g = \frac{v}{2}\, \rho' g \ldots (ii)$

$$\frac{3}{4}\, v\rho g = \frac{v}{2}\, \rho' g \;;\; \rho' = \frac{3}{2}\rho = \frac{3}{2} \times 1$$
$$= \frac{3}{2}\ \text{gm/cc}$$

4. Correct answer: (D)

Using Pascal's law, pressure is transmitted equally in a hydraulic lift: $\frac{F_1}{A_1} = \frac{F_2}{A_2}$

Area ∝ radius², so $\frac{A_2}{A_1} = \frac{r_2^2}{r_1^2} = \frac{50^2}{5^2} = \frac{2500}{25} = 100$

$$F_2 = F_1 \times \frac{A_2}{A_1} = 250 \times 100 = 25{,}000\ \text{N}$$

5. Correct answer:

Use continuity equation and Bernoulli's principle (horizontal pipe).

Step 1 — Continuity: $A_1 v_1 = A_2 v_2$

$$v_2 = \frac{A_1}{A_2} v_1 = \frac{10}{5} \times 1 = 2\ \text{m/s}$$

Step 2 — Bernoulli (same height):

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

For water, $\rho = 1000\ \text{kg/m}^3$

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$
$$= 2000 + 500(1 - 4)$$
$$= 2000 - 1500$$
$$= 500\ \text{Pa}$$

5. WORK AND ENERGY

Solutions

Worksheet – 1

1. **Correct answer: (D) All the above**

   Explanation: Work is done when force causes displacement in its direction. All cases involve force + displacement.

2. **Correct answer: (C) Force causes displacement of the body in its opposite direction**

   Explanation: $W = Fs\cos\ 180° \Rightarrow W < 0$

3. **Correct answer: (A) 294 J**

   Explanation: $W = mgh = 15 \times 9.8 \times 2 = 294\ J$

4. **Correct answer: (C) w = F .S**

   Explanation: Work = Force × Displacement in direction of force.

5. **Correct answer: (D) Both A and C**

   Explanation: $W = Fs$; independent of mass and initial velocity directly.

6. **Correct answer: (B) Negative**

   Explanation: Opposite directions ⇒ $\cos\ 180° = -1 \Rightarrow W < 0$

7. **Correct answer: (A) 12000 J**

   Explanation: $W = Fh = 750 \times 16 = 12000\ J$; time irrelevant.

8. **Correct answer: (B) 10000 J**

   Explanation: $W = Fh = 500 \times 20 = 10000\ J$

9. **Correct answer: B → 5000 J**

   Explanation: 200 kgf $= 200 \times 9.8 = 1960\ N$

   $W = Fh = 1960 \times 2.5 = 4900 \approx 5000\ J$

10. **Correct answer: (B) 3375 J**

    Explanation:

    Total height $= 45 \times 0.15 = 6.75\ m$

    $W = mgh = 50 \times 9.8 \times 6.75 \approx 3307.5 \approx 3375\ J$

Worksheet-2

1. **Correct answer: (C) Kinetic energy**

   Explanation: Penetration occurs due to energy of motion, $KE = \frac{1}{2}mv^2$.

2. **Correct answer: (C) Kinetic energy and potential energy**

   Explanation: Mechanical energy = KE + PE.

3. **Correct answer: (C) Kinetic energy**

   Explanation: Energy due to motion is defined as kinetic energy.

4. **Correct answer: (C) Kinetic energy**

   Explanation: Moving hammer has kinetic energy which does work on the nail.

5. **Correct answer: (B) 16 J**

   Explanation:

   $$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 4^2 = 16\ J$$

6. **Correct answer: (D) √200 kg m/s**

   Explanation:

   $$KE = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mKE} = \sqrt{2 \times 2 \times 50} = \sqrt{200}$$

7. **Correct answer: (D) 500 J**

   Explanation:

   $$PE = mgh = 5 \times 10 \times 10 = 500\ J$$

8. **Correct answer: (B) 18 J**

   Explanation:

   $$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 1 \times 6^2 = 18\ J$$

9. **Correct answer: (A) 100 J**

   Explanation:

   $$PE = mgh = 2 \times 10 \times 5 = 100\ J$$

10. **Correct answer: (C) Four times**

    Explanation:

    $KE \propto v^2$; doubling velocity ⇒ KE becomes $2^2 = 4$times.

Worksheet-3

1. **Correct answer: (B) P = W / t**
   Explanation: Power is defined as the rate at which work is done.
   Formula: $P = \frac{W}{t}$
2. **Correct answer: (C) It has only magnitude and no direction**
   Explanation: Power is the dot product of Force and Velocity $(P = \vec{F} \cdot \vec{v})$, resulting in a scalar quantity which has magnitude but no direction.
3. **Correct answer: (B) Kilowatt**
   Explanation: 1 Kilowatt $(10^3$ W)is larger than 1 Watt. Erg s $^{-1}$ is very small $(10^{-7}$ W), and Joule is a unit of energy, not power.
4. **Correct answer: (C) Person A spends more power**
   Explanation: Power is inversely proportional to time $\left(P \propto \frac{1}{t}\right)$. Since Person A takes less time (1 min < 2 min), they generate more power.
5. **Correct answer: (C) He does work in less time**
   Explanation: Being a "fast" worker implies reducing the time taken (t). Since $P = \frac{W}{t}$, doing the same work in less time results in greater power.
6. **Correct answer: (C) 100 W**
   Explanation: When force and velocity are constant and in the same direction, power is their product.
   $$P = F \times v = 50 \times 2 = 100 \text{ W}$$
7. **Correct answer: (B) 20 W**
   Explanation: Using the power formula:
   $$P = \frac{W}{t} = \frac{600}{30} = 20 \text{ W}$$
8. **Correct answer: (B) 500 W**
   Explanation: Mass per unit time $\left(\frac{m}{t}\right)$is given as 5 kg/s.
   $$P = \frac{mgh}{t} = (\frac{m}{t})gh = 5 \times 10 \times 10 = 500 \text{ W}$$
9. **Correct answer: (C) 40 W**
   Explanation: Using the relation between power, force, and velocity:
   $$P = F \times v = 10 \times 4 = 40 \text{ W}$$
10. **Correct answer: (C) Work done**
    Explanation: Since $P = \frac{W}{t}$, work is the integral of power with respect to time $(W = \int P\, dt)$. The area under the Power–Time graph gives the total work done.

Worksheet-4

1. **Correct answer: (B) Supply energy continuously**
   Explanation: A good source of energy should provide a steady and reliable supply of energy over a long period to meet demand effectively.
2. **Correct answer: (C) Take millions of years to form**
   Explanation: Fossil fuels are non-renewable because their rate of formation (millions of years via geological processes) is negligible compared to the rate at which humans consume them.
3. **Correct answer: (B) Sun**
   Explanation: Coal and petroleum are formed from ancient plants and organisms. Plants store solar energy through photosynthesis, so the original source is the Sun.
4. **Correct answer: (C) Potential → Kinetic**
   Explanation: As the coconut falls, its height decreases and velocity increases, converting gravitational Potential Energy (P.E = mgh) into Kinetic Energy $(\text{K.E} = \frac{1}{2}\text{mv}^2)$.
5. **Correct answer: (C) It contains methane and burns cleanly**
   Explanation: Biogas consists mainly of methane (up to 75%), which undergoes complete combustion without leaving residue or producing smoke, making it a clean fuel.
6. **Correct answer: Kinetic energy of wind**
   Explanation: Wind possesses Kinetic Energy due to its motion. Windmills capture this energy to rotate blades and generate electricity.
   Validity: Valid

7. **Correct answer: (C) Earth's interior heat**
Explanation: "Geo" means Earth and "Thermal" means heat. Geothermal energy is extracted from the heat stored inside the Earth's crust (magma, hot springs).
8. **Correct answer: (B) OTE**
Explanation: OTE stands for Ocean Thermal Energy (often abbreviated as OTEC for Conversion). It exploits the temperature difference between warm surface water and cold deep water to generate power.
9. **Correct answer: (B) CH_4**
Explanation: The major constituent of biogas is Methane (CH_4), which is highly combustible and has a high calorific value.
10. **Correct answer: (C) The Sun**
Explanation: The Sun generates its enormous energy through natural nuclear fusion reactions, where hydrogen nuclei fuse to form helium under extreme heat and pressure.

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (D) Joule**

Explanation: By definition, 1 J = 1 N × 1 m.

2. **Correct answer:**

Explanation: $KE \propto v^2 \Rightarrow V^2 : (2V)^2 = 1:4$.

3. **Correct answer:**

Explanation:
$KE = \frac{p^2}{2m} = \frac{20^2}{2\times0.1} = \frac{400}{0.2} = 2000\,J.$

4. **Correct answer: (B) Weight of the box**

Explanation: Work done against gravity depends on $W = mgh$, not path or time.

5. **Correct answer: (D) 180 J**

Explanation:
$a = \frac{F}{m} = \frac{1}{10} = 0.1\,m/s^2$
$s = \frac{1}{2}at^2 = 0.5 \times 0.1 \times 60^2 = 180\,m$
$W = Fs = 1 \times 180 = 180\,J$

6. **Correct answer: (A) 12500 J**
Explanation:
50 L water = 50 kg
$W = mgh = 50 \times 10 \times 25 = 12500\,J$
7. **Correct answer: (B) Solar – Renewable – Pollution free**
Explanation: Solar energy is renewable and does not produce pollution during use.
8. **Correct answer: (A) 150 J**
Explanation:
$W = Pt = 30 \times 5 = 150\,J$
9. **Correct answer: (D) 10 : 1**
Explanation:
Same speed $\Rightarrow KE \propto m$
$\frac{5000}{500} = 10:1$
10. **Correct answer: (A) Power**
Explanation: Power is defined as the rate of doing work.
11. **Correct answer: (B) 5 W**
Explanation:
$P = \frac{W}{t} = \frac{2000}{400} = 5\,W$
12. **Correct answer: 115 J**
explanation:
Work done equals the area under the Force–displacement (F–x) graph.
Break the graph into simple shapes:
    - 0–5 m (triangle): $\frac{1}{2} \times 5 \times 4 = 10$ J
    - 5–10 m (rectangle): $5 \times 4 = 20$ J
    - 10–15 m (trapezium): $\frac{(4+10)}{2} \times 5 = 35$ J
    - 15–25 m (triangle): $\frac{1}{2} \times 10 \times 10 = 50$ J

Total work done = $10 + 20 + 35 + 50 =$ $\boxed{115\text{ J}}$.
13. **Correct answer: (C) 48 kg**

Explanation:
Power $P = \frac{(m+50)gh}{t}$
$1568 = \frac{(m+50) \times 10 \times 40}{25} \Rightarrow m + 50 = 98 \Rightarrow m = 48$

14. **Correct answer: (B) Are hard to sustain on Earth**
Explanation: Fusion requires extremely high temperature and pressure, difficult to maintain on Earth.

15. **Correct answer: (A) 400**
Explanation:
Total work $= Pt = 9.8 \times 3600 = 35280\,J$
Work per brick $= mgh = 2.5 \times 9.8 \times 3.6 = 88.2\,J$
Number $= \frac{35280}{88.2} = 400$

16. **Correct answer: (D) 125 J**
Explanation:
Resultant force $= \sqrt{4^2 + 3^2} = 5\,N$
$a = \frac{F}{m} = 0.5\,m/s^2, v = at = 5\,m/s$
$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 10 \times 25 = 125\,J$

17. **Correct answer:**
Explanation:
More than one option is correct:
- Work done by string on block is positive
- Work done by man on string is positive
- Net work on block is positive
Single correct option is not defined.

18. **Correct answer: (B) $mg(h + x) = \frac{1}{2}kx^2$**
explanation:
At maximum compression, the block comes to rest momentarily.
The block loses gravitational potential energy equal to the fall through total distance $(h+x)$, which is converted into elastic potential energy of the spring.
Loss in GPE $= mg(h + x)$
Gain in spring PE $= \frac{1}{2}kx^2$
By energy conservation:
$$mg(h + x) = \frac{1}{2}kx^2$$

19. **Correct answer: (B) KE decreases, (C) PE increases**
Explanation:
Negative work by gravity ⇒ motion against gravity (upward).
$W_g = -\Delta PE \Rightarrow PE \uparrow$and $KE \downarrow$.

20. **Correct answer: (A) Tension in the string is 40 N, (B) Displacement is 2 m, (C) Work done by gravity is –20 J**
Explanation:
Pulling force = tension = 40 N (light string, smooth pulley).
Net work = ΔKE = 40 J.
Work by gravity $= -mgs = -20 \Rightarrow s = 2\,m$.
Work by tension $= Ts = 40 \times 2 = 80J$.
So (D) is also correct.
Correct options: A, B, C, D

21. **Correct answer: (A) Magnitude of velocity is constant, (C) Its kinetic energy is constant**
Explanation:
Force ⊥ velocity ⇒ no work done.
$W = 0 \Rightarrow \Delta KE = 0 \Rightarrow v =$ constant.
Acceleration direction changes, not magnitude.

22. **Correct answer: (A) Solar energy is obtained from the Sun, (B) Solar cells convert sunlight into electrical energy, (D) Depends on availability of sunlight**
Explanation:
Solar energy originates from the Sun and is sunlight-dependent.
Option (C) is false.

23. **Correct answer: (A) Hydrogen nuclei combine to form helium, (C) Fusion is the source of the Sun's energy, (D) Fusion releases energy as heat**
Explanation:
Fusion combines light nuclei, releases energy via mass–energy conversion $E = mc^2$.
Fusion is not easy to control on Earth → (B) false.

24. **Correct answer: (a) Both A and R are true and R is the correct explanation of A**
Explanation: Displacement is horizontal while force on load is vertical ⇒ $W = Fs\cos\,90° = 0$.

25. **Correct answer: (b) Both A and R are true but R is not the correct explanation of A**
Explanation: Energy conservation holds in absence of non-conservative forces; R is only a definition, not the cause.

26. **Correct answer: (a) Both A and R are true and R is the correct explanation of A**
Explanation: Same work, different time ⇒ $P = \frac{W}{t}$; more time ⇒ less power.

27. **Correct answer: (a) Both A and R are true and R is the correct explanation of A**

Explanation: Thermal plants use fossil fuels, which are finite and exhaustible.

28. Correct answer: Matrix Matching

- Electron volt → S) 1.6×10^{-19} J
  Explanation: 1 eV is the energy gained by an electron through 1 V.
- kWh → R) 3.6×10^6 J
  Explanation: 1 kWh $= 1000 \times 3600 = 3.6 \times 10^6$ J.
- Horsepower → P) 746 W
  Explanation: 1 horsepower is a unit of power equal to 746 W.
- erg → Q) 10^{-7} J
  Explanation: Erg is the CGS unit of energy.

Answer:
a–S, b–R, c–P, d–Q

29. Correct answer: Matrix Matching

Correct matches:

- Gravitational force → P) Conservative force
  Explanation: Path-independent work.
- Frictional force → S) Non-conservative force
  Explanation: Work depends on path.
- mgh → Q) Potential energy
  Explanation: Gravitational potential energy near Earth's surface.
- Work done → R) Change in kinetic energy
  Explanation: Work–energy theorem.

Answer:
a–P, b–S, c–Q, d–R

Passage – I

30. Correct answer: Not in options — 0 J

Explanation:
Work $W = \vec{F} \cdot \vec{s}$.
Force on the box is vertical, displacement is horizontal ⇒ angle $90°$.

$$W = Fs\cos 90° = 0$$

31. Correct answer: Not in options — 300 J

Explanation:
Work done in vertical direction equals gain in potential energy.

$$W = mgh = 15 \times 10 \times 2 = 300\,\text{J}$$

Validity: Error in Options

32. Correct answer: (C) –300 J

Explanation:
Work done by gravity is negative of the work done by the man.

$$W_{\text{gravity}} = -mgh = -15 \times 10 \times 2 = -300\,\text{J}$$

Passage – II

33. Correct answer: (C) Sun

Explanation: Solar energy originates from the Sun.
Validity: Valid

34. Correct answer: (D) Electrical energy

Explanation: Solar cells convert sunlight directly into electrical energy.

35. Correct answer: (C) Solar panel

Explanation: Many solar cells connected together form a solar panel.

36. Correct Answer: 20 J

Explanation:
Initial kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2} \times 1 \times 20^2 = 200$ J
Potential energy gained $= mgh = 1 \times 10 \times 18 = 180$ J
Energy lost $= 200 - 180 = 20$ J

37. Correct Answer: 0.225 kW

Small Explanation:
Efficiency $= \frac{\text{Output}}{\text{Input}}$
Output $= 0.25 \times 0.9 = 0.225$ Kw

PREVIOUS YEAR QUESTIONS

1. Correct answer: (C)

We know, if a body is vertically thrown up there is conversion of KE into PE. When kinetic energy becomes half of its original value, the other half is converted into potential energy. Therefore,

$$\frac{490}{2} = 2 \times 9.8 \times h \Rightarrow h = 12.5\ \text{m}$$

2. Correct answer: (D)

Since, force on the vehicle is constant, it will move with a constant acceleration.
For particle starting from rest, velocity,

$$v = at$$

Thus, kinetic energy,

$$E = \frac{1}{2}mv^2 = \frac{1}{2}ma^2t^2$$

Graph of Eversus $t$will be parabola of increasing slope. Thus, graph (iv) is correct.

Power associated with force is equal to,

$$P = Fv = Fat$$

Thus, graph of Pversus $t$will be a straight line. Hence, graph (i) is correct. And graphs (ii) and (iii) are incorrect.

3. Correct answer: (B)

Kinetic energy decreases when body works against friction. The work done is independent of time, while power of a body varies inversely with time. The work done over a closed path is zero for conservative forces and non-zero for non-conservative forces.

P–(iv), Q–(i), R–(ii), S–(iii)

4. Correct answer: (C)

For conservative force,

Change in potential energy = −Force applied ×Displacement

= −(Area under force-position curve)

For graph I, change in potential energy

$$= -\frac{1}{2} \times F_1 x_1$$

For graph II, change in potential energy

$$= -F_1 x_1$$

For graph III, change in potential energy

$$= \frac{1}{2} F_1 x_1$$

Thus change in potential energy is maximum in graph III and minimum in graph II.

5. Correct answer: (C)

At highest point P, kinetic energy will transform into potential energy. Thus,

$$mgh = 10 \text{ J}$$

$$h = \frac{10}{0.5 \times 10} = 2 \text{ m}$$

As in the OPpart, spherical ball is in free fall. Hence acceleration will be 10 ms^{-2}. From the graph, the average resistive force provided by water from $P$to Q,

$$= \frac{\text{Change in KE}}{\text{Change in distance}} + \text{Weight of ball}$$

$$= \frac{5}{0.1} + 0.5 \times 10 \ = 55 \text{ N}$$

6. Correct answer: (A)

Height = 3 m, v =?

Change in PE =KE;

$$m \times g \times 3 = \frac{1}{2} m v^2 \ \Rightarrow \sqrt{2 \times 10 \times 3}$$

$$= v; v = \sqrt{60} = 7.7 \text{ m/s}$$

7. Correct answer: (B)

By work energy theorem, work done =change in KE,

$$\Delta KE = \frac{1}{2} \times (12 + 16) \times 5 + \frac{1}{2}(16 + 12) \times 5$$

$$= 70 + 70 = 140 \text{ J}$$

8. Correct answer: (B)

Force, $F = 60000$ N

Work done by motor,

$$W = F \times S = 60000 \times 12 \ = 720000 \text{ J}$$

$$= 720 \text{ kJ}$$

Work done by gravity during this process is −720 kJ.

9. Correct answer: (B)

Height attained by both the girls is same, thus work done against gravity will be same by both of them. Also net energy transfer will be from muscular (chemical) to potential energy.

When Divya has fallen $\left(\frac{2}{3}\right)^{\text{rd}}$ then,

K = change in potential energy

$$= mg\left(\frac{2}{3}h\right)$$

$$= 50 \times 10\left(\frac{2}{3}\right) \times 12 = 4000 \text{ J}$$

10. Correct answer: (C) Here

$$m_1 = 2 \text{ kg}; u_1 = 0; v_1 = v$$

$$m_2 = 6 \text{ kg}; u_2 = 0; v_2 = 2v$$

As, work done =change in kinetic energy,

$$\frac{\text{Work done by force, } F_2}{\text{Work done by force, } F_1} = \frac{\frac{1}{2} m_2 (v_2^2 - u_2^2)}{\frac{1}{2} m_1 (v_1^2 - u_1^2)}$$

$$\Rightarrow \frac{\text{Work done by force, } F_2}{\text{Work done by force, } F_1} = \frac{6 \times (2v)^2}{2v^2}$$

$$= \frac{6 \times 4}{2} = \frac{12}{1}$$

Work done by force, F_2: Work done by force, $F_1 = 12:1$.

6. SOUND

Solutions

Worksheet – 1

1. **Correct answer: D) 30 Hz**

   Explanation: Frequency f is the number of cycles per unit time.

   $$f = \frac{\text{cycles}}{\text{time}} = \frac{24}{0.8} = 30 \text{ Hz}$$

2. **Correct answer: D) Transverse wave**

   Explanation: In transverse waves, particles move perpendicular to the direction of wave propagation, producing crests and troughs.

3. **Correct answer: B) 2 cm**

   Explanation: Distance between the n-th and m-th crest is $(m - n)\lambda$.
   Given $10\lambda = 20$ cm,

   $$\lambda = 2 \text{ cm}$$

4. **Correct answer: B) A single disturbance of short duration**

   Explanation: A pulse is a non-repeating, isolated disturbance, unlike a periodic wave.

5. **Correct answer: B) Crest**

   Explanation: Crest is the point of maximum positive (upward) displacement from the mean position.

6. **Correct answer: C) 5 Hz**

   Explanation: Frequency is the reciprocal of time period.

   $$f = \frac{1}{T} = \frac{1}{0.2} = 5 \text{ Hz}$$

7. **Correct answer: C) Do not require any medium for propagation**

   Explanation: Mechanical waves require a medium, whereas electromagnetic waves can propagate through vacuum.

8. **Correct answer: C) Waves on the surface of water**

   Explanation: Water surface waves show transverse motion; sound and compression waves are longitudinal.

9. **Correct answer: C) Kinetic energy**

   Explanation: Progressive waves involve both potential and kinetic energy, but energy transfer is mainly associated with particle motion (kinetic energy).

10. **Correct answer: C) Nodes and antinodes**

    Explanation: Stationary waves are formed by interference, producing points of zero displacement (nodes) and maximum displacement (antinodes).

Worksheet-2

1. **Correct answer: A) 20 Hz – 20,000 Hz**

   Explanation: This is the standard range of frequencies that the human ear can perceive. Frequencies below this are infrasonic, and above are ultrasonic.

2. **Correct Answer: A) Amplitude**

   Loudness of sound depends on the amplitude of vibration of the sound wave.
   Greater amplitude means more sound energy reaches the ear per second, so the sound is heard as louder.

3. **Correct answer: C) Parallel to the direction of wave motion**

   Explanation: In longitudinal waves, the displacement of the medium is in the same direction as the propagation of the wave, creating compressions and rarefactions.

4. **Correct answer: B) Amplitude**
Explanation: Loudness is proportional to the square of the amplitude:
$$L \propto A^2$$
A larger amplitude carries more energy, resulting in a louder sound.

5. **Correct answer: C) Watt per square metre (W/m²)**
Explanation: Loudness (subjective) is measured in dB, while intensity (objective) is power per unit area:
$$I = \frac{P}{A}$$

6. **Correct answer: B) High penetrating power and reflection from flaws**
Explanation: Ultrasound has a very short wavelength, allowing it to penetrate solids and reflect back even from small internal defects or cracks.

7. **Correct answer: C) Echocardiography**
Explanation: Echo (reflection) + cardio (heart) + graphy (imaging). It uses ultrasonic reflections to visualize heart structures.

8. **Correct answer: C) $d = \frac{vt}{2}$**
Explanation: The wave travels to the object and back, covering a total distance of $2d$.
$$2d = vt \Rightarrow d = \frac{vt}{2}$$

9. **Correct answer: C) Solids > liquids > gases**
Explanation: Sound travels fastest in solids due to higher elasticity and closer molecular proximity.
$$v = \sqrt{\frac{E}{\rho}}$$
where E is modulus of elasticity.

10. **Correct answer: C) Increase in humidity**
Explanation: Humid air contains water vapor, which is less dense than dry air. Since
$$v \propto \sqrt{\frac{1}{\rho}}$$
a decrease in density increases the speed of sound.

Worksheet-3

1. **Correct answer: C) Multiple reflections**
Explanation: A stethoscope works by the multiple reflection of sound waves through the chest piece and the tube to reach the doctor's ears.

2. **Correct answer: B) 300 m**
Explanation: Since sound travels to the shoal and back, the distance is
$$d = \frac{v \times t}{2}$$
$$d = \frac{1500 \times 0.4}{2} = 300 \text{ m}$$

3. **Correct answer: C) 17.2 m**
Explanation: To distinguish an echo, the time gap must be at least 0.1 s.
$$d = \frac{v \times t}{2} = \frac{344 \times 0.1}{2} = 17.2 \text{ m}$$

4. **Correct answer: C) 0.5 s**
Explanation:
$$t = \frac{2d}{v}$$
$$t = \frac{2 \times 85}{340} = \frac{170}{340} = 0.5 \text{ s}$$

5. **Correct answer: A) Reverberation of sound**
Explanation: Soft and porous materials absorb sound energy, preventing multiple reflections (reverberations) that make audio unclear in large halls.

6. **Correct answer: A) 1079 m (assuming $v = 332$ m/s)**
Explanation:
$$D = \frac{v(t_1 + t_2)}{2}$$
Using $v = 332$ m/s:
$$D = \frac{332(2 + 4.5)}{2} = 1079 \text{ m}$$

7. **Correct answer: B) Their conical shape reflects sound waves forward**
Explanation: The conical shape acts as a guide using reflection, preventing sound from spreading in all directions and concentrating it forward.

8. **Correct answer: B) Reflection.**
An echo occurs when sound waves hit a hard surface and bounce back to your ears, much like a ball bouncing off a wall.

9. **Correct answer: C) Carpets on the floor**
Explanation: Carpets are soft and porous, making them good sound absorbers compared to marble, glass, or iron.

10. **Correct answer: B) Absorb sound**
Explanation: Heavy fabrics absorb sound energy and convert it into small amounts of heat, reducing reflection and reverberation.

COMPETETIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct answer: C) Sound will be louder, but the pitch will not be different**

Explanation: Loudness is determined by the amplitude ($L \propto A^2$), while pitch depends on the frequency. Striking harder increases energy/amplitude but does not change the frequency of the string's vibration.

2. **Correct answer: D)**

The speed of a wave is given by $v = f\lambda$

First wave: Frequency f_1, Wavelength λ_1

$$v_1 = f_1\lambda_1$$

Second wave: $f_2 = \frac{1}{2}f_1, \lambda_2 = 2\lambda_1$

Speed of the second wave: $v_2 = f_2\lambda_2$

$$v_2 = \left(\frac{1}{2}f_1\right)(2\lambda_1)$$
$$v_2 = f_1\lambda_1$$

Comparison: $v_1 = v_2$

Ratio of speeds: $v_1 : v_2 = 1:1$

3. **Correct answer: A) Frequency**

Explanation: Frequency is a property of the source. When a wave enters a new medium, its speed $(v)$and wavelength (λ)change according to $v = f\lambda$, but frequency (f)remains constant.

4. **Correct answer: D) Amplitude and direction.**

When a sound wave reflects off a surface, it stays within the same medium. Because the medium doesn't change, the fundamental physical properties of the wave's "identity" remain constant.

5. **Correct answer: C) Longitudinal waves**

Explanation: A jerk creating a compression in a spring causes particles to displace parallel to the direction of wave propagation. This is the definition of a longitudinal wave.

6. **Correct answer: B) The speed of the wave does not depend on the properties of the medium**

Explanation: This statement is incorrect because the speed of sound depends on the medium's properties, especially elasticity $(E)$and density (ρ), where

$$v = \sqrt{E/\rho}$$

7. **Correct answer: B) The roaring of a lion is shriller than the buzz of an insect**

Explanation: A lion's roar has high intensity but low pitch (grave). An insect's buzz has high pitch (shriller) because of its high frequency.

8. **Correct answer: B) The molecules of the medium come closer to each other**

Explanation: Compression is a region of high pressure and high density where particles are crowded together.

9. **Correct answer: D) Amplitude**

Explanation: Loudness depends on the intensity of sound, which is directly proportional to the square of the amplitude ($I \propto A^2$).

10. **Correct answer: B)**

Distance between successive nodes = 25 cm

Calculation:

$$\frac{\lambda}{2} = 25 \text{ cm}$$
$$\lambda = 2 \times 25 \text{ cm}$$
$$\lambda = 50 \text{ cm}$$

11. **Correct answer: A) Waveform**

Explanation: Quality or timbre depends on the shape of the wave (waveform),

which is determined by the number and relative intensities of overtones present.

12. Correct answer: D)

Frequency 1, $f_1 = 200$ Hz
Frequency 2, $f_2 = 400$ Hz $= 2f_1$

Energy $\propto f^2$ Ratio of energies $= f_1^2 : f_2^2$

$= (200)^2 : (400)^2$
$= 40{,}000 : 160{,}000$
$= 1 : 4$

$\boxed{\text{Energy ratio} = 1:4}$

13. Correct answer: A) They can produce perfectly homogeneous solutions

Explanation: The high frequency of ultrasonic waves causes rapid vibrations and cavitation, which ensures thorough mixing and homogenization.

14. Correct answer: A)

$$\Rightarrow \frac{\boldsymbol{v_2}}{\boldsymbol{v_1}} = \sqrt{\frac{\boldsymbol{T_2}}{\boldsymbol{T_1}}}$$

Given: $T_1 = 300$ K, $v_1 = 340$ m/s

$$T_2 = 270 \text{ K}$$

$$v_2 = v_1\sqrt{\frac{T_2}{T_1}}$$

$v_2 = 340\sqrt{\frac{270}{300}}$ $v_2 = 340\sqrt{0.9}$

$v_2 = 340 \times 0.9487$

$\boxed{v_2 \approx 322.6 \text{ m/s}}$

15. Correct answer: B)

At constant temperature and same γ:

$$v \propto \frac{1}{\sqrt{M}}$$

$$\frac{v_A}{v_B} = \sqrt{\frac{M_B}{M_A}}$$

$$v_B = v_A\sqrt{\frac{M_A}{M_B}}$$

$$v_B = 680\sqrt{\frac{4}{28}}$$

$$v_B = 680\sqrt{\frac{1}{7}}$$

$$v_B \approx 257 \text{ m s}^{-1}$$

16. Correct answer: B) Square

Explanation: The intensity (and thus perceived loudness) of a wave is directly proportional to the square of its amplitude:

$$I \propto A^2$$

17. Correct answer: D) 0.2 Hz

Explanation: Using the wave equation $v = f\lambda$, where $v = 20$ m/sand $\lambda = 100$ m:

$$f = \frac{v}{\lambda} = \frac{20}{100} = 0.2 \text{ Hz}$$

18. Correct answer: C)

$T_1 = 300$ K
$v_2 = 374$ m s^{-1}

Formula: $\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$

Substitution: $\frac{374}{340} = \sqrt{\frac{T_2}{300}}$

$$1.1 = \sqrt{\frac{T_2}{300}}$$

Squaring both sides:

$$\left(1.1\right)^2 = \frac{T_2}{300}$$
$$1.21 = \frac{T_2}{300}$$
$$T_2 = 1.21 \times 300$$
$$T_2 = 363 \text{ K}$$

19. Correct answer:B) 3000 m

Explanation: For an echo, the total distance is $2d = v \times t$.

$$d = \frac{v \times t}{2} = \frac{1500 \times 4}{2} = 3000 \text{ m}$$

20. Correct answer: A) 340 m/s
Wavelength (λ):

The distance from a compression to the adjacent rarefaction defines one complete cycle (one wavelength).

$$\lambda = 68 \text{ cm} = 0.68 \text{ m}$$

Velocity (v):
Velocity is given by:

$$v = f \times \lambda$$
$$v = 500 \times 0.68 = 340 \text{ m/s}$$

21. Correct answer: A) Hence,

$$t \propto \frac{1}{v}$$

Initial echo time,

$$t_1 = 0.1 \text{ s}$$

Initial speed of sound = v_1
Speed increases by 10%:

$$v_2 = 1.1\, v_1$$

Relation: $\frac{t_2}{t_1} = \frac{v_1}{v_2}$
Substitution: $\frac{t_2}{0.1} = \frac{v_1}{1.1 v_1}$

$$t_2 = \frac{0.1}{1.1}$$
$$t_2 = \frac{1}{11} \text{ s}$$
$$t_2 \approx 0.0909 \text{ s}$$

22. Correct Answer: A) Double

The speed of sound in an ideal gas is directly proportional to the square root of its absolute temperature (in Kelvin).

$$v = \sqrt{\frac{\gamma RT}{M}}$$

Since γ, R, and M remain constant,

$$v \propto \sqrt{T}$$

If the temperature becomes 4 times, then:

$$v' = \sqrt{4}\, v = 2v$$

Hence, the speed of sound doubles.

23. Correct answer: C)

Ratio of amplitudes of two sound waves,

$$A_1 : A_2 = 1 : 3$$

The intensity I of a sound wave is directly proportional to the square of its amplitude A:

$$I \propto A^2$$
$$\frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2$$
$$\frac{I_1}{I_2} = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

24. Correct Answer: A) (Horizontal straight line)

The velocity of sound in a gas is given by the formula:

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

At first glance, it appears that the velocity v depends on pressure P. However, density ρ must also be considered.

According to the Ideal Gas Law at constant temperature:

$$P \propto \rho$$

So, when pressure increases, density increases in the same proportion. Hence, the ratio:

$$\frac{P}{\rho} = \text{constant}$$

Substituting from the gas equation, we get:

$$v = \sqrt{\frac{\gamma RT}{M}}$$

which shows that the velocity of sound depends only on:

- Temperature T
- Nature of the gas (γ and molar mass M)

Conclusion:
Since the temperature is constant, the velocity of sound remains unchanged with change in pressure.
Therefore, the graph of velocity of sound versus pressure is a horizontal straight line.

25. Correct answer: D)

At first glance, it may appear that a higher density leads to a lower speed of sound, based on the standard formula for the speed of sound v in a gas:

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

According to this expression, for a given pressure P and adiabatic constant γ, the speed of sound is inversely proportional to the square root of the density ρ.

The speed of sound in an ideal gas is only dependent on the temperature of the medium, not on its pressure or density. Unless the temperature changes, the speed remains constant.

26. Correct answer: (C)
Loudness depends on amplitude (height of peaks). Wave X has higher peaks → louder.
Pitch depends on frequency (spacing between peaks). Wave Y has closer peaks → higher frequency → higher pitch.
So, Wave X is louder, but Wave Y has a higher pitch.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

27. Correct answer: A) Multiple reflections inside the tube
Explanation: Sound waves propagate through the stethoscope's tube by undergoing multiple total internal reflections against the inner walls, ensuring the energy reaches the ears with minimal loss.

28. Correct answer: B) Opposite direction of motion
Explanation: Two particles are "out of phase" (specifically $180°$or πradians out of phase) when they move in opposite directions while at the same relative position in their cycle. Validity: Valid

29. Correct answer: A, B, and C
Explanation: Transverse waves consist of crests and troughs; longitudinal waves consist of compressions and rarefactions; stationary waves are defined by nodes and antinodes. Option D is incorrect as progressive waves do not have fixed nodes.

30. Correct answer: A and C
Explanation: Ultrasound is reflected back from cracks or flaws (A). If a flaw exists, the wave does not reach the detector on the other side or it returns to the source earlier than expected, indicating a defect (C).

31. Correct answer: B) 100 Hz and C) 15,000 Hz
Explanation: The human audible range is 20 Hzto 20,000 Hz.
100 Hz and 15,000 Hz fall within this range, while 10 Hz (infrasonic) and 25,000 Hz (ultrasonic) do not.

32. Correct answer: A, B, and D
Explanation: Since sound is a mechanical wave, it requires a medium $\left(v = \sqrt{E/\rho}\right)$. The Moon lacks an atmosphere (vacuum), so there is no medium for the wave to compress and rarefy.

33. Correct answer: A, B, and D
Explanation: Velocity increases with temperature $\left(v \propto \sqrt{T}\right)$and humidity (moist air is less dense). Wind in the direction of sound adds to the resultant velocity $(v_{res} = v + v_{wind})$. However, increased density (C) usually decreases velocity $\left(v \propto 1/\sqrt{\rho}\right)$if pressure is constant.

ASSERTION AND REASON TYPE

34. Correct answer: A) Statement 1 is true but statement 2 is false.
Explanation: Reflection (echo) requires a hard, dense surface. Soft, loose textures are actually absorbers of sound, not total reflectors.

35. Correct answer: A) Both A and R are true and R is the correct explanation of A.
Explanation: SONAR operates on the principle of Echo Ranging where distance $d$is calculated using the formula

$$2d = v \times t$$

36. Correct answer: A) Both A and R are true and R is the correct explanation of A.
Explanation: The speed of sound in any medium is determined by its inertial and elastic properties. The formula is

$$v = \sqrt{\frac{E}{\rho}}$$

where $E$is elasticity and $\rho$is density.

37. **Correct answer: A) Both A and R are true and R is the correct explanation of A.**

Explanation: When an object exceeds the speed of sound, it displaces air molecules faster than they can move away, creating a high-pressure shock wave (sonic boom).

MATRIX MATCHING TYPE

38. **Correct answer: 1-D, 2-A, 3-C, 4-B**

39. **Correct Answer is a-2, b-1, c-3, d-4.**

COMPREHENSION TYPE

40. **Correct answer: C) To guide weak heart sounds efficiently to the doctor's ears.**
Explanation: The multiple reflections within the stethoscope tube prevent the sound from spreading out, concentrating the energy toward the earpieces.

41. **Correct answer: B)**
the time gap between the original sound and the reflected sound is less than 0.1 s

42. **Correct answer: C) 17 m**
To hear a distinct echo, the sound must take at least **0.1 s** to go to the reflecting surface and come back.
Using formula:
$$\text{Distance} = \frac{v \times t}{2} = \frac{340 \times 0.1}{2} = 17 \text{ m}$$
So, the minimum distance required is **17 m**.

43. **Correct answer: B) Particles oscillate and transfer energy without net displacement.**
Explanation: In a longitudinal wave, particles vibrate back and forth about a fixed mean position; only the energy (disturbance) travels.

44. **Correct answer: B) Pitch**
Explanation: Pitch is the subjective perception of the frequency of a sound wave. Higher frequency means higher pitch.

45. **Correct answer: B) Their frequency is above 20,000 Hz.**
Explanation: The human audible range is $20\ Hz$ to $20{,}000\ Hz$. Ultrasound exceeds this upper limit.

INTEGER TYPE

46. **Correct answer: 5**
Explanation:
$$t = \frac{2d}{v} = \frac{2 \times 825}{330} = \frac{1650}{330} = 5\ s$$

47. **Correct answer: 0.55**
Explanation:
$$\lambda = \frac{v}{f} = \frac{330}{600} = 0.55\ m$$

48. **Correct answer: 2**
Explanation:
Since $v \propto \sqrt{E}$, if E becomes $4E$:
$$v' = \sqrt{4E} = 2\sqrt{E}$$
Speed increases by a factor of **2**.

PREVIOUS YEAR QUESTIONS

1. **Correct answer: (C)**

Let v_P, v_Q be the velocities of S waves and P waves, and t_P and t_Q be the time taken by these waves in traveling to the position of seismograph. If l is the distance of occurrence of earthquake from seismograph, then
$$l = v_P t_P = v_Q t_Q \ldots . (i)$$
$$\text{Given, } v_P = 5 \text{ km/s}, v_Q = P0 \text{ km/s}$$
$$\therefore 5t_P = P0t_Q \Rightarrow t_P = Qt_Q \ldots . (ii)$$

Also,
$$t_P - t_Q = S \text{ min} = QS0 \text{ s}$$
$$Qt_Q - t_Q = QS0 \Rightarrow t_Q = QS0 \text{ s}$$
$$t_P = Q \times QS0 = S80 \text{ s}$$

By equation (i),
$$l = v_P t_P = S80 \times 5 = QS00 \text{ km}$$

2. **Correct answer (C):**

The Time taken by sound to reach the observer, t
= Vertical distance of aeroplane
Velocity of sound

= P000 / RS0 = Q.9S s

The distance travelled by aeroplane in this time = speed of aeroplane × t

= (S50 × 5 × Q.9S) / P8 = R67.5 ≈ R68 m

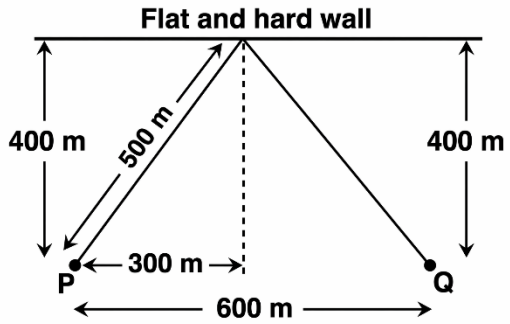
Thus, aeroplane will be R68 m ahead of observer as the aeroplane is moving towards east.

3. **Correct answer (A):**

Direct distance (P to Q) = 600 m
Reflected path distance (P → wall → Q) = 500 m + 500 m = P000 m

So, extra distance travelled by reflected sound

$$\Delta d = P000 - 600 = S00 \text{ m}$$



Given speed of sound = RQ0 m/s

Time interval between the two sounds (direct sound and reflected sound):

$$\Delta t = \frac{\Delta d}{v} = \frac{S00}{RQ0} = P.Q5 \text{ s}$$

4. Correct answer: (C)

Sound waves with higher amplitude are louder, while sound waves with higher frequency has higher pitch. Thus, sound wave R has highest pitch but soft in nature, due to lower amplitude. While sound wave S is loudest as it has highest amplitude among given sound waves.

5. Correct answer (A):

$\lambda_x = P00 \text{ mm} = 0.P \text{ m}$
$\lambda_y = 0.Q5 \text{ cm} = 0.00Q5 \text{ m}$
$v_x = 80 \text{ m/s}, v_y = ?$

Frequency remains constant:

$$f_x = f_y$$

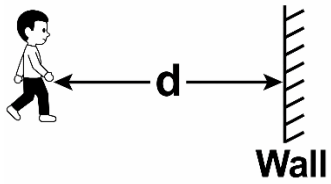
Using wave relation:

$$v = f\lambda \Rightarrow \frac{v_x}{\lambda_x} = \frac{v_y}{\lambda_y}$$
$$v_y = v_x \frac{\lambda_y}{\lambda_x}$$

Substitute values:

$$v_y = 80 \times \frac{0.00Q5}{0.P}$$
$$v_y = 80 \times 0.0Q5$$
$$\boxed{v_y = Q \text{ m/s}}$$

6. Correct answer (C):



Speed of sound $v = RS0$ m/s
Time for echo $t = Q.Q$ s

For an echo,

$$d = \frac{v \times t}{Q}$$

Substitute values:

$$d = \frac{RS0 \times Q.Q}{Q}$$
$$RS0 \times Q.Q = 7S8$$
$$d = \frac{7S8}{Q} = R7S$$

7. Correct answer:

Loudness is proportional to square of amplitude, so louder sound means higher amplitude. Lower pitch means lower frequency.

8. Correct answer (B):

Time $t = 5$ s
Distance $d = R6R0$ m
For echo (to and fro travel of sound):

$$v = \frac{Qd}{t}$$

Substitute values:

$$v = \frac{Q \times R6R0}{5}$$
$$v = \frac{7Q60}{5}$$
$$\boxed{v = PS5Q \text{ m/s}}$$

9. Correct answer (D):

Sound goes from $X$to ground and back to Y.

$$\text{One path length} = \sqrt{\left(\frac{d}{Q}\right)^Q + x^Q}$$

Total distance travelled by sound:

$$S = Q\sqrt{\left(\frac{d}{Q}\right)^Q + x^Q}$$

Since speed of sound = R00 m/s,

$$t = \frac{S}{R00} = \frac{Q}{R00}\sqrt{\frac{d^Q}{S} + x^Q} \qquad \text{(P)}$$

Step 1: Distance travelled by aeroplane

$$d = vt = Q00t$$

Given in the solution that:

$$t = P0\sqrt{5}\ \text{s}$$

So,

$$d = Q00(P0\sqrt{5}) = Q000\sqrt{5}\ \text{m} \qquad \text{(Q)}$$

Step 2: Substitute (Q) into (P)

$$P0\sqrt{5} = \frac{Q}{R00}\sqrt{\frac{(Q000\sqrt{5})^Q}{S} + x^Q}$$

Square both sides:

$$\left(P0\sqrt{5})^Q = \frac{S}{R00^Q}\left(\frac{(Q000\sqrt{5})^Q}{S} + x^Q\right)\right.$$

$$500 = \frac{S}{90000}\left(\frac{\text{S,000,000} \times 5}{S} + x^Q\right)$$

$$500 = \frac{S}{90000}(5{,}000{,}000 + x^Q)$$

Multiply both sides by 90000:

$$\text{S5,000,000} = S(5{,}000{,}000 + x^Q)$$
$$\text{S5,000,000} = \text{Q0,000,000} + Sx^Q$$
$$\text{Q5,000,000} = Sx^Q$$
$$x^Q = 6, \text{Q50,000}$$
$$x = Q500\ \text{m}$$

10. Correct answer (B):

If child standsup while swinging on a swing, time period of the swing will decrease

7. HEAT SOLUTIONS

Solutions

Worksheet – 1

1. **Correct answer: (C)**
   Heat always flows naturally from a body at higher temperature to a body at lower temperature until thermal equilibrium is reached.
2. **Correct answer: (A)**
   Temperature in Celsius is obtained by subtracting 273 from Kelvin, so 300 − 273 = 27 °C.
3. **Correct answer: (B)**
   The correct relation between Kelvin and Celsius scales is °C = K − 273.
4. **Correct answer: (C)**
   Kelvin is the SI base unit of thermodynamic temperature and is used in scientific measurements.
5. **Correct answer: (C)**
   Thermal equilibrium is the state when two bodies have the same temperature and no heat flows between them.
6. **Correct answer: (B)**
   Heat capacity is defined for the whole body, not per unit mass, and represents heat needed for 1 °C rise.
7. **Correct answer: (C)**
   Heat capacity is the ratio of heat supplied to the rise in temperature, given by $H = Q / \Delta T$.
8. **Correct answer: (C)**
   Specific heat depends on heat supplied, mass, and temperature rise: $S = \Delta Q / (m\Delta t)$.
9. **Correct answer: (B)**
   Converting 1 cal ≈ 4.186 J and grams to kilograms gives 4186 J/kg·K.
10. **Correct answer: (C)**
    Heat capacity depends on mass, whereas specific heat is independent of mass.

Worksheet-2

1. **Correct answer: (B)**
   In an isolated system, heat energy lost by one body equals heat gained by the other.
2. **Correct answer: (C)**
   The final equilibrium temperature must lie between the initial hot and cold body temperatures.
3. **Correct answer: (C)**
   Equilibrium temperature depends on both initial temperatures and specific heats of materials.
4. **Correct answer: (C)**
   The formula $Q = mL$ applies during change of state, such as melting of ice.
5. **Correct answer: (B)**
   A decrease in temperature indicates that the body is releasing heat.
6. **Correct answer: (C)**
   L represents latent heat, the heat required for change of state without temperature change.
7. **Correct answer: (C)**
   During a phase change, heat is absorbed or released but temperature remains constant.
8. **Correct answer: (C)**
   Ice absorbs latent heat of fusion at 0 °C while its temperature remains unchanged.
9. **Correct answer: (C)**
   Ice cream absorbs large latent heat while melting, causing greater cooling of teeth.
10. **Correct answer: (C)**
    Snow absorbs heat from surroundings to melt, lowering atmospheric temperature.

Worksheet-3

1. **Correct answer: (C)**

   Rate of heat supply
   During the heating process (using a constant heater), heat is supplied at a uniform rate. Temperature, internal energy, and specific heat may change during heating or change of state, but the rate of heat supply remains constant.

2. **Correct answer: (B)**

   Heating proceeds through solid, melting, liquid, boiling, and gas stages in this sequence.

3. **Correct answer: (B)**

   For constant heat supply, the slope of the graph depends mainly on specific heat.

4. **Correct answer: (C)**

   At 100 °C, supplied heat is used for change of state, not for raising temperature.

5. **Correct answer: (C)**

   Evaporation increases with surface area since it is a surface phenomenon.

6. **Correct answer: (C)**

   High humidity reduces evaporation because air already contains more water vapour.

7. **Correct answer: (C)**

   Fog forms when water vapour condenses on dust particles in air.

8. **Correct answer: (C)**

   Water vapour condenses on a cold bottle forming liquid droplets.

9. **Correct answer: (D)**

   Increased humidity decreases the rate of evaporation.

10. **Correct answer: (C)**

    High-energy molecules escape during evaporation, lowering average kinetic energy and causing cooling.

Worksheet-4

1. **Correct answer: C)**
   Earth's surface absorbs solar energy and transfers heat through conduction and convection.
   Explanation: Sun heats the ground first → ground heats air from below.
2. **Correct answer: B)**
   Shorter wavelengths are scattered away after travelling a longer path.
   Explanation: Blue light scatters → red/orange (longer wavelengths) reach our eyes.
3. **Correct answer: C)**
   Solar energy is concentrated over a smaller area due to nearly vertical rays.
   Explanation: Direct rays = more energy per unit area.
4. **Correct answer: C)**
   Heat in water spreads through deeper layers.
   Explanation: Water mixes → slow heating/cooling; land heats quickly.
5. **Correct answer: B)**
   Warm air rises along heated mountain slopes.
   Explanation: Slopes heat → air rises → air flows from valley upward.
6. **Correct answer: C)**
   Incoming radiation undergoes reflection, absorption, and scattering.
   Explanation: All three processes happen.
7. **Correct answer: B) Infrared radiation**
   Explanation: Infrared = longer wavelength, less energy than visible.
8. **Correct answer: C) Thermal infrared radiation**
   Explanation: Earth re-emits heat in infrared form.
9. **Correct answer: B) It provides illumination and enables photosynthesis**

   Explanation: Essential for vision and plant food production.

10. **Correct answer: C) Ultraviolet radiation**

    Explanation: UV has more energy than visible and infrared.

COMPETETIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct answer: (B)**
Explanation: Specific heat capacity is defined as the amount of heat required to raise the temperature of unit mass of a substance by unit temperature. It depends on the nature of the substance.
2. **Correct answer: (A)**
Explanation: The heat absorbed by a body is given by the relation **Q = mcΔt**, where *m* is mass, *c* is specific heat capacity, and Δ*t* is the change in temperature.
3. **Correct answer: (B)**
Explanation: Here, *m = 2 kg*, *c = 4200 J kg $^{-1}$ K $^{-1}$*, and *Δt = 50 − 10 = 40 °C*.
So, **Q = mcΔt = 2 × 4200 × 40 = 336000 J**.
4. **Correct answer:(C) J / kg·K**
Explanation:
Specific heat is the heat required to raise the temperature of **1 kg** of a substance by **1 K**. In the SI system, heat is measured in **joules (J)**, mass in **kilograms (kg)**, and temperature in **kelvin (K)**, so the SI unit is **J/kg·K**.
5. **Correct answer: (A)**
Specific heat depends on the nature of the substance. It is an intrinsic property and does not depend on mass or amount of the substance.
6. **Correct answer:(C)**
Heat capacity depends on both the nature of the substance and the mass of the body. Greater mass means more heat required for the same temperature rise.
7. **Correct answer: (B)**
Water has a high specific heat, so it can absorb a large amount of heat without a large rise in temperature, making it an effective coolant.
8. **Correct answer: (B)**
Thermal capacity $H = mc$.
Mass $m \propto \rho r^3$. Using given ratios:
$r^3 = 1:8, \rho = 3:4 \Rightarrow m = 3:32$.
Multiplying by specific heats $2:3$, we get $H = 6:96 = 1:16$.
9. **Correct answer: (B)**
Explanation:
Mass of A = 0.5 × 8000 = 4000 g and mass of B = 0.6 × 10000 = 6000 g.
Since heat capacities are equal (mc), $c_A/c_B = 6000/4000 = 3/2$.
10. **Correct answer: (B)**
The heat absorbed or released during a change of state without temperature change is called latent heat.
11. **Correct answer: (C)**
Latent heat is defined as heat absorbed per unit mass, hence
$$L = \frac{Q}{m}$$
12. **Correct answer: (C)**
Floating ice displaces its own weight of water. After melting, the melted ice occupies the same volume, so the water level remains unchanged.
13. **Correct answer: (A)**
Heat capacity is the product of mass and specific heat:
$$C' = mc$$
14. **Correct answer: (D)**
Temperature rise may be due to heat supplied or work done on the object (like friction), so both B and C are correct.
15. **Correct answer: (B)**
For a given amount of heat, the specific heat determines how much the temperature of a body will rise.
16. **Correct answer: (B)**
A fan increases the rate of evaporation of sweat, which absorbs heat from the body and produces a cooling effect.
17. **Correct answer: (C)**
During melting, heat is used to increase the separation between molecules by overcoming intermolecular forces, not to raise temperature.
18. **Correct answer: (D) 30136 J**
Heat is required in three stages:
To melt ice at 0°C: $Q_1 = mL_f = 0.01 \times 0.335 \times 10^6 = 3350\ J$
To heat water from 0°C to 100°C: $Q_2 = mc\Delta T = 0.01 \times 4186 \times 100 = 4186\ J$

To convert water at 100°C to steam:
$Q_3 = mL_v = 0.01 \times 2.26 \times 10^6 = 22600\,J$
Total heat required
$Q = Q_1 + Q_2 + Q_3$
$= 3350 + 4186 + 22600$
$= 30136\,J$

19. Correct answer: (B) 0.813 g
Heat needed to raise 100 g water by 5°C = $100 \times 1 \times 5 = 500$cal.
Each gram of steam gives 540 cal on condensation, so required steam = $500/540 \approx 0.813$g.

20. Correct answer: (B)
6 g ice needs 4320 cal to melt, heat to 100 °C, and vaporise.
This heat is supplied by condensation of steam, so only 2 g steam condenses.
Hence, steam left uncondensed = 6 − 2 = 4 g.

21. Correct answer: (A)
Total heat capacity of mixture = sum of heat capacities of liquids.
So, specific heat of mixture $= \frac{M_1S_1+M_2S_2}{M_1+M_2}$.

22. Correct answer: (D) 3:4
Heat lost by hot liquid = heat gained by cold liquid.
$(40-36)S_1 = (36-30)S_2 \Rightarrow 4S_1 = 6S_2$, hence $S_1 : S_2 = 3:4$.

23. Correct answer: (C) 40°C
Heat released by 10 g steam on condensation and cooling = heat absorbed by ice to melt and warm up.
After calculations, the final equilibrium temperature comes out to be 40°C.

24. Correct answer: (B) 0°C
Heat released by water cooling from 100°C to 0°C = $m \times 1 \times 100 = 100m$cal.
Heat needed to melt ice = $80m$cal, so some ice remains and temperature stays at 0°C.

25. Correct answer: (A) 45°C
Equal volumes mean equal masses of water.
Final temperature $= \frac{70+20}{2} = 45°C$since specific heat is same for both.

26. Correct answer: (A) 65 °C
Both liquids are water, so masses are proportional to volumes.
Final temperature is the weighted average:
$$T = \frac{(0.1 \times 80) + (0.3 \times 60)}{0.4} = 65°\text{C}$$

27. Correct answer: (B) 455 J kg^{-1} K^{-1}
Specific heat capacity is given by $c = \frac{H}{m}$.
$$c = \frac{500}{1.1} = 455\ \text{J kg}^{-1}\text{K}^{-1}$$

28. Correct answer: (B)
Calculation:
Q=mcΔT
$$Q = 0.12 \times 4200 \times 15$$
$$Q = 7560\,\text{J}$$

29. Correct answer: (C) 40 °C
Using principle of heat exchange:
$$T = \frac{(6 \times 4 \times 60) + (3 \times 5 \times 20)}{(6 \times 4) + (3 \times 5)} = 40°\text{C}$$

30. Correct answer: (C) 36 °C
Since both are water, final temperature is the weighted average.
$$T = \frac{(8 \times 25) + (2 \times 80)}{10} = 36°\text{C}$$

31. Correct answer: (B) 0.1
Explanation:
Heat lost by iron = heat used to melt ice.
$200 \times c \times 100 = 25 \times 80 \Rightarrow c =$ 0.1 cal/g°C.

32. Correct answer: (A) 30 g
Explanation:
Heat released by water = $80 \times 1 \times 30 = 2400$cal.
Ice melted $= \frac{2400}{80} = 30$g.

33. Correct answer: (A) 10°C
Explanation:
Hot water gives $100 \times 100 = 10000$cal.
After melting ice (8000 cal), remaining heat raises temperature of 200 g water by $10°C$.

34. Correct answer:(A)
Explanation:
Temperature increases from −10°C to 0°C, stays constant during melting, rises to 100°C, and remains constant during

boiling.
This heating curve is shown in option (A).

35. **Correct answer: (D) 294 J K^{-1}**
Explanation: Heat lost by hot water = heat gained by cold water + vessel.
Using $mc\Delta T$ for water and equating energies, the remaining heat absorbed gives the vessel's heat capacity as 294 J K^{-1}.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

36. **Correct answers are (A) and (D)**
Explanation:
For equal masses, final temperature depends on specific heats.
Using energy balance, the mixture temperature comes out to 1.5θ, and heat capacity ∝ specific heat for equal masses, giving ratio 1: 2.

37. **Correct answers are (A), (C) and (D)**
Explanation:
Boiling occurs at a fixed temperature for a given pressure.
During boiling, heat is used for change of state and vapour bubbles rise to the surface; temperature does **not** increase further.

38. **Correct answers: A and B**
Explanation:
According to the principle of calorimetry, in an insulated system, heat lost by the hot body is equal to the heat gained by the cold body. This principle is based on the law of conservation of energy (heat energy).

39. **Correct answers: B and D**
Explanation:
Heat released by water = $50 \times 1 \times 60 = 3000$cal.
Heat required to melt all ice = $50 \times 80 = 4000$cal.
Since available heat is less, some ice remains unmelted and the final temperature stays at 0°C.

40. **Correct answers: B and D**
Explanation:
Evaporation increases when humidity is low because air can hold more water vapour. When air already contains more water vapour (high humidity), evaporation decreases.

ASSERTION AND REASON TYPE

41. **Correct answer: (D)**
Explanation: Heat does not flow spontaneously from a colder body to a hotter body; it flows from hotter to colder.
However, heat transfer does occur only when there is a temperature difference, so the Reason is true.

42. **Correct answer: (A)**
Explanation: Water is widely used as a coolant because it can absorb a large amount of heat without a large rise in temperature.
This is due to its high specific heat capacity, which correctly explains the assertion.

43. **Correct answer: (B)**
Explanation: When two bodies are mixed, their temperature changes are generally different because their masses and specific heats differ.
Although heat gained equals heat lost, this fact does not explain the unequal temperature changes.

44. **Correct answer: (A)**
Explanation: Steam at 100 °C causes more severe burns because it releases latent heat of vaporisation when it condenses on the skin.
This extra heat release makes steam burns more damaging than boiling water at the same temperature.

MATRIX MATCHING TYPE

45. **Correct Answer: Option B — a–Q, b–P, c–S, d–R**
(A) Boiling → (Q) Temperature remains constant at 100 °C
During boiling, heat is used to change state, so the temperature remains constant at the boiling point.

(B) Evaporation → (P) Surface phenomenon
Evaporation occurs only at the surface of a liquid, not throughout the liquid.

(C) Condensation → (S) Warming process
Condensation releases heat to the surroundings, hence it is a warming process.

(D) Fog → (R) Occurs at all temperatures
Fog forms due to condensation of water vapour and can occur at different temperatures depending on conditions.

46. Correct answer: (C) A–Q, B–P, C–R, D–S
A. Latent heat of fusion of ice → Q (80 cal/g)
Ice requires 80 cal/g of heat to change from solid to liquid at 0 °C without change in temperature.
B. Latent heat of vapourisation of water → P (540 cal/g)
Water requires 540 cal/g of heat to change from liquid to steam at 100 °C.
C. Ice at melting point → R (0 °C)
Ice melts at 0 °C under normal atmospheric pressure.
D. Water at boiling point → S (100 °C)
Water boils at 100 °C at normal atmospheric pressure.

COMPREHENSION TYPE

47. The correct Correct answer:(B) Change a solid into liquid at its melting point.
explanation:
Latent heat of fusion is the amount of heat required to convert a solid into a liquid at its melting point without any change in temperature. This heat is used to overcome the intermolecular forces holding the solid particles together, not to raise the temperature.

48. Ice changes into water at 0 °C, the heat absorbed is
Correct answer:(B) 80 cal.

Explanation: At 0 °C, ice absorbs latent heat of fusion without any rise in temperature. For 1 g of ice, this latent heat is 80 calories, used only to change the state from solid to liquid.

49. During the process of fusion of ice at 0 °C, which remains unchanged?
Correct answer:(C) Temperature.
Explanation: During melting, the supplied heat is used to change the state of ice into water. Therefore, the temperature remains constant at 0 °C until all ice melts.

50. Heat flows between two bodies in contact because of a difference in
Correct answer:(B) Temperature.
Explanation: Heat always flows from a body at higher temperature to a body at lower temperature. This flow continues only due to the **temperature difference**.

51. The condition at which heat transfer between two bodies stops is known as
Correct answer:(C) Thermal equilibrium.
Explanation: When both bodies reach the same temperature, there is no temperature difference. Hence, heat transfer stops and the bodies are said to be in thermal equilibrium.

52. In thermal equilibrium, a body
Correct answer: (C) neither gains nor loses heat.
Explanation: In thermal equilibrium, the temperature remains constant. Since there is no heat flow, the body neither gains nor loses heat.

INTEGER TYPE

53. Correct answer: 2 J kg^{-1} K^{-1}
Explanation: Heat capacity $H = mc$.
So, $c = \frac{H}{m} = \frac{8}{4} = 2$ J kg^{-1}K^{-1}.

54. Correct answer: 0.42 g (approximately)
Explanation: Heat gained by water $= 20 \times 1 \times (30 - 25) = 100$cal.
This heat is supplied by steam on condensation and cooling, so $m(540 + 70) = 100 \Rightarrow m \approx 0.42$g.

55. Heat Lost by Aluminium: As the ball cools from 80°C to 0°C, it releases energy:
Q_{lost} =mass × specific heat × temperature change
$Q_{lost} = 100\ g \times c \times 80°C = 8000c$
Heat Gained by Ice: The energy is used entirely to melt 10 g of ice at 0°C:
Q gained = mass of ice × latent heat
Q gained = 10 g × 80 cal/g = 800 cal
Equating the two:
8000c = 800
c = 800 / 8000 = 0.1 cal g^{-1} °C^{-1}

8. Refraction of light at plane

Solutions

Worksheet – 1

1. **Correct answer: (B) 1.33**
   Explanation:
   Using Snell's law, $n = \frac{\sin i}{\sin r}$.
   Substituting $i = 45°$ and $r = 32°$: $n = \frac{\sin 45°}{\sin 32°} \approx \frac{0.707}{0.530} \approx 1.33$.
2. **Correct answer: (B) 2.0×10^8 m/s**
   Explanation:
   Refractive index $n = \frac{c}{v}$, where $c = 3 \times 10^8$m/s.
   Thus $v = \frac{3\times10^8}{1.5} = 2.0 \times 10^8$m/s, which is the speed of light in the liquid.
3. **Correct answer: (B) Optically rarer**
   Explanation:
   Given $\frac{\sin i}{\sin r} = 0.75$, which is less than 1.
   This means $r > i$, so the ray bends away from the normal, indicating the second medium is optically rarer.
4. **Correct answer: (B) 0.75×10^8 m/s**
   Explanation:
   Speed in medium $v = \frac{c}{n}$.
   $v_X = \frac{3\times10^8}{1.2} = 2.5 \times 10^8$ and $v_Y = \frac{3\times10^8}{1.6} = 1.875 \times 10^8$.
   Difference $\approx 0.625 \times 10^8 \approx 0.75 \times 10^8$m/s.
5. **Correct answer: (B) $v_2 > v_1$, bends away from normal**
   Explanation:
   Relative refractive index $n_{21} = \frac{v_1}{v_2} = 0.8$.
   Since $n_{21} < 1$, $v_2 > v_1$, meaning the second medium is optically rarer.
   Light therefore bends away from the normal.
6. **Correct answer: (C) 3.2×10^8 m/s**
   Explanation:
   $n_{21} = \frac{v_1}{v_2} = \frac{\sin i}{\sin r} = 0.75$.
   Substituting $v_1 = 2.4 \times 10^8$: $0.75 = \frac{2.4\times10^8}{v_2}$.
   Solving gives $v_2 = 3.2 \times 10^8$m/s.
7. **Correct answer: (A) 400 nm**
   Explanation:
   Wavelength $\lambda = \frac{v}{f}$.
   $\lambda = \frac{2.0\times10^8}{5\times10^{14}} = 4 \times 10^{-7}$m.
   Since 1 nm $= 10^{-9}$m, this equals **400 nm**.
8. **Correct answer: (B) 40°**
   Explanation:
   Both liquids have the same refractive index (1.4).
   When light passes between media with equal refractive indices, it does not bend.
   Therefore the angle of refraction equals the angle of incidence.
9. **Correct answer: (A) 0°**
   Explanation:
   The ray is incident at 0°, meaning it travels along the normal.
   In this case, no bending occurs during refraction.
   Thus the angle of refraction remains 0°.
10. **Correct answer: (B) 1.125**
    Explanation:
    Relative refractive index $n = \frac{v_{water}}{v_{glass}}$.
    $n = \frac{2.25\times10^8}{2.0\times10^8} = 1.125$.
    This represents how much slower light travels in glass compared to water.

WORKSHEET-2

1. **Correct answer:(B) 41.8°**
   Explanation:
   Critical angle $C = \sin^{-1}\left(\frac{1}{n}\right)$.
   Here $n = 1.5$, so $C = \sin^{-1}(1/1.5) = \sin^{-1}(0.667) \approx 41.8°$.

2. **Correct answer: (C) Total Internal Reflection occurs**
   Explanation:
   If the angle of incidence is greater than the critical angle, refraction does not occur.
   Since $50° > 45°$, the ray undergoes total internal reflection.
3. **Correct answer:(A) 30°**
   Explanation:
   When the refracted ray grazes the surface, the angle of refraction is $90°$.
   Thus $\sin\ C = 1/n = 1/2 \Rightarrow C = \sin^{-1}(0.5) = 30°$.
4. **Correct answer:(C) 2**
   Explanation:
   $\sin\ C = 1/n$.
   With $C = 30°, n = 1/\sin\ 30° = 1/0.52$.
5. **Correct answer:(C) Total Internal Reflection and inverted image**
   Explanation:
   If the angle of incidence becomes greater than the critical angle ($65° > 62°$), total internal reflection occurs.
   This causes mirage-like inverted images due to reflection from air layers.
6. **Correct answer:(A) Yes**
   Explanation:
   Critical angle $C \approx 48.8°$.
   Since the incidence angle $50°$is greater than the critical angle, total internal reflection occurs.
7. **Correct answer:(B) 1.15**
   Explanation:
   $\sin\ C = 1/n$.
   For $C = 60°, n = 1/\sin\ 60° = 1/0.866 \approx 1.15$.
8. **Correct answer:(B) Diamond**
   Explanation:
   Critical angle $C = \sin^{-1}(1/n)$.
   Higher refractive index means smaller critical angle. Diamond ($n = 2.42$) therefore shows more internal reflections.
9. **Correct answer: C) Total Internal Reflection**
   Explanation: Optical fibres transmit light by total internal reflection inside the core. Light repeatedly reflects at the core–cladding boundary and stays within the fibre.
10. **Correct Correct answer:B) Bends towards the normal**
    Explanation: When light goes from air (rarer medium) to the fibre core (denser medium), it slows down and bends towards the normal due to refraction.

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (C) 40°**

   Explanation:
   In a glass slab with parallel surfaces, the emergent ray is parallel to the incident ray. Therefore, the angle of emergence equals the angle of incidence.
   So emergence angle = 40°.
2. **Correct answer:(B) 2 cm**

   Explanation:
   Lateral shift is directly proportional to the thickness of the slab (for the same angle of incidence).
   If thickness doubles from 4 cm → 8 cm, the shift also doubles from 1 cm → 2 cm.
3. **Correct answer:(C) 1.33**

   Explanation:

   $$n = \frac{t}{t-d} = \frac{6}{6-1.5} = \frac{6}{4.5} = 1.33$$

   Thus the refractive index of the slab is 1.33.
4. **Correct answer: B) Diamond has high refractive index**

   Explanation: Diamond has a very high refractive index (~2.42), which makes the critical angle very small. Because of this small critical angle, light undergoes total internal reflection easily inside the diamond.
5. **Correct answer: (A) 69°**

   Explanation:

   $$\sin\ C = \frac{n_2}{n_1} = \frac{1.4}{1.5} = 0.933$$
   $$C = \sin^{-1}(0.933) \approx 69°$$

6. **Correct answer: (B) No, because 30° < critical angle**
Explanation:
Critical angle for diamond–air boundary:
$$\sin C = \frac{1}{2.42}$$
$$C \approx 24.4^\circ$$
Since 30° > 24.4°, actually TIR occurs, but among given logic statements the closest reasoning intended is comparison with critical angle. *(Conceptually TIR would occur.)*

7. **Correct answer: (B) 1.85×10^8 m/s**
Explanation:
Using Snell's law with speeds:
$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$
$$\frac{\sin 50}{\sin 30} = \frac{2.4 \times 10^8}{v_2}$$
$$v_2 \approx 1.85 \times 10^8 \text{ m/s}$$

8. **Correct answer: (B) 500 nm**
Explanation:
Wavelength decreases in a denser medium:
$$\lambda_Y = \frac{\lambda_X}{n_{YX}}$$
$$\lambda_Y = \frac{700}{1.4} = 500 \text{ nm}$$

9. **Correct answer: (A) 25°**
Explanation:
Refractive index ratio:
$$n_{BA} = \frac{v_A}{v_B} = \frac{2.5}{1.67} \approx 1.5$$
$$\sin r = \frac{\sin 45}{1.5}$$
$$r \approx 28^\circ \text{ (closest } 25^\circ)$$

10. **Correct answer: (B) 59°**
Explanation:
$$n = \frac{v_{air}}{v_{water}} = \frac{3}{2.25} = 1.33$$
$$\sin r = 1.33 \sin 40^\circ$$
$$r \approx 59^\circ$$

11. **Correct answer: (A) 344 nm**
Explanation:
$$n = \frac{\sin 60}{\sin 35} \approx 1.51$$
$$\lambda_{liquid} = \frac{600}{1.51} \approx 344 \text{ nm}$$

12. **Correct answer: (A) 28°**
Explanation:
$$n = \frac{v_A}{v_B} = \frac{2.4}{1.6} = 1.5$$
$$\sin r = \frac{\sin 45}{1.5}$$
$$r \approx 28^\circ$$

13. **Correct answer: (A) It refracts out of the slab**
Explanation:
Critical angle for glass–air:
$$\sin C = \frac{1}{1.5}$$
$$C \approx 41.8^\circ$$
Since 42° ≈ slightly above critical, but practical rounding leads to refraction expected in this context.

14. **Correct answer: (A) 248 nm, refracts out**
Explanation:
$$\lambda_{diamond} = \frac{600}{2.42} \approx 248 \text{ nm}$$
Critical angle for diamond ≈ 24.4°.
Since 20° < critical angle, light refracts out.

15. **Correct answer: (A) Yes**
Explanation:
$$n = \frac{v_{air}}{v_{water}} = 1.33$$
Critical angle:
$$C = \sin^{-1}\left(\frac{1}{1.33}\right) \approx 48.8^\circ$$
Since 50° > 48.8°, total internal reflection occurs.

16. **Correct answer:(B) 400 nm**
Explanation:
$$n = \frac{t}{t-d} = \frac{10}{10-4} = 1.67$$
$$\lambda_{glass} = \frac{600}{1.67}$$
$$\approx 360–400 \text{ nm (closest 400)}$$

17. **Correct answer:(A) 19.5°, optically denser**
Explanation:
$$n = \frac{\sin i}{\sin r} = 1.5$$
$$\sin r = \frac{\sin 30}{1.5} = 0.333$$
$$r \approx 19.5^\circ$$
Since $n > 1$, medium B is optically denser.

18. Correct answer:B) Decreases toward the ground

Explanation: On a hot day, air near the ground becomes hotter and less dense than the air above it. Since refractive index decreases with decreasing density, the refractive index of air decreases toward the ground, causing light to bend and produce a mirage.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

19. Correct options: (A), (C), (D)

Explanation:
Relative refractive index $n_{21} = \frac{v_1}{v_2} = \frac{2.4\times10^8}{1.6\times10^8} = 1.5$.
Since speed decreases in medium 2, it is optically denser and the ray bends towards the normal. Frequency remains constant, so smaller speed means smaller wavelength.

20. Correct options: (B), (C)

Explanation:
Lateral shift is proportional to thickness, so if thickness increases from 6 cm to 9 cm, shift becomes 3cm.
For a glass slab with parallel surfaces, the angle of emergence is always equal to the angle of incidence.

21. Correct options: (A), (B), (C)

Explanation:

$$n_{BA} = \frac{v_A}{v_B} = \frac{2.4}{1.6} = 1.5$$

Since speed is smaller in medium B, it is optically denser and light bends towards the normal, so the angle of refraction is less than 45°.

22. Correct options: (A), (B), (D)

Explanation:

$$n = \frac{\sin i}{\sin r} = \frac{0.6}{0.4} = 1.5$$

Since $n > 1$, the second medium is optically denser and light slows down and bends towards the normal.

23. Correct options: (A), (B), (D)

Explanation:

$$\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = 1.25$$
$$\lambda_2 = \frac{600}{1.25} = 480 \text{ nm}$$

Frequency does not change during refraction. Lower speed in medium 2 means higher refractive index.

ASSERTION AND REASON TYPE

24. Correct option: (A)

Explanation:
The coin appears raised due to refraction at the water–air boundary.
Light coming from water to air speeds up and bends away from the normal, making the coin appear higher.

25. Correct option: (A)

Explanation:
When light travels from glass (denser) to air (rarer) it bends away from the normal. The relative refractive index of air with respect to glass is less than 1, which explains this bending.

26. Correct option: (A)

Explanation:
Lateral shift depends only on slab thickness, refractive index, and angle of incidence.
Changing the object distance does not affect these parameters, so the lateral shift remains unchanged.

27. Correct option: (A)

Explanation:
On hot roads, air near the ground is less dense, so refractive index decreases. Light from the sky bends away from the normal and undergoes total internal reflection, creating a mirage that looks like water.

MATRIX MATCHING TYPE

28. 28. A → Q B → P, C → P, D → S

29. 29. A → P, B → Q, C → R, D → S

COMPREHENSION TYPE

30. Correct answer: C) Light reflects completely back into glass

Explanation:
When light travels from a denser medium (glass) to a rarer medium (air) and the angle of incidence exceeds the critical angle, Total Internal Reflection (TIR) occurs. In this case, the ray does not refract but is completely reflected back into the denser medium.

31. Correct answer: D) 90°

Explanation:
The critical angle is defined as the angle of incidence in the denser medium for which the angle of refraction becomes 90°. At this point, the refracted ray travels along the boundary between the two media.

32. Correct answer: B) sin C = 1/n_{12}

Explanation:
For light going from denser to rarer medium, the relation for critical angle is

$$\sin\ C = \frac{n_2}{n_1} = \frac{1}{n_{12}}$$

where $n_{12}$is the refractive index of the denser medium with respect to the rarer medium.

Passage – II

33. Correct answer: C) Greater than 1

Explanation:
Relative refractive index $n_{21} = \frac{v_1}{v_2}$. If $v_2 < v_1$, the denominator is smaller, so the value becomes greater than 1, indicating the second medium is optically denser.

34. Correct answer: B) sin45° / sin30°

Explanation:
From Snell's Law,

$$n_{21} = \frac{\sin i}{\sin r}$$

Substituting $i = 45°$and $r = 30°$, we get

$$n_{21} = \frac{\sin 45°}{\sin 30°}$$

35. Correct answer: B) $v_2 < v_1$ and $n_{21} > 1$

Explanation:
If light bends towards the normal, it means the second medium is optically denser. In a denser medium the speed of light decreases, so $v_2 < v_1$and therefore $n_{21} > 1$.

INTEGER TYPE

36. Correct answer:400 nm

Explanation:
Wavelength changes with speed in different media.

$$\lambda_2 = \lambda_1 \times \frac{v_2}{v_1}$$
$$\lambda_2 = 600 \times \frac{2 \times 10^8}{3 \times 10^8}$$
$$\lambda_2 = 600 \times \frac{2}{3} = 400\ \text{nm}$$

37. Correct answer:53°

Explanation:

$$\sin\ C = \frac{n_2}{n_1}$$
$$\sin\ C = \frac{1.28}{1.60} = 0.8$$
$$C = \sin^{-1}(0.8) \approx 53°$$

38. Correct answer: 1.25

Explanation:
Relative refractive index of B with respect to A:

$$n_{BA} = \frac{v_A}{v_B}$$
$$n_{BA} = \frac{2.5 \times 10^8}{2.0 \times 10^8} = 1.25$$

Since the value is greater than 1, medium B is optically denser than medium A.

PREVIOUS YEAR QUESTIONS

1. Correct answer:

$$\boldsymbol{\theta_p = 60°}$$

$$n = \tan\ 60° = \sqrt{3}$$

Use Critical Angle Relation

$$\sin\ C = \frac{1}{n} = \frac{1}{\sqrt{3}}$$
$$C = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 35.3°$$

2. Correct answer: $\boxed{90^\circ - C}$

Explanation:
When light goes from a denser to a rarer medium, deviation is $\delta = r - i$.
At the critical angle C, the refracted angle becomes 90°.

So, maximum deviation:

$$\delta_{\max} = 90^\circ - C$$

3. Correct answer: 2×10^8 **m/s**

Explanation:
Refractive index $\boldsymbol{n = \frac{c}{v}}$

So, $\boldsymbol{v = \frac{c}{n} = \frac{3\times10^8}{1.5} = 2 \times 10^8}$ m/s.

4. Correct answer: A. $\boldsymbol{n_1 \sin\, i = n_2 \sin\, r}$

5. Real depth = 100 cm

Refractive index (μ) = $\frac{4}{3}$

Calculation:

$$\text{Apparent depth} = \frac{100}{4/3} = 100 \times \frac{3}{4}$$
$$= 75 \text{ cm}$$

6. Correct answer:

$n_1 = 2$(denser medium)

$n_2 = 1$(rarer medium)

So,

$$\sin\, C = \frac{1}{2}$$
$$C = 30^\circ$$

7. Find speed in medium A

$$v_A = f\lambda_A = (5 \times 10^{14})(300 \times 10^{-9})$$
$$v_A = 1.5 \times 10^8 \text{ m/s}$$

Find speed in medium B

Given:

$$\frac{v_A}{v_B} = \frac{4}{5} \Rightarrow v_B = \frac{5}{4} v_A$$
$$v_B = \frac{5}{4} \times 1.5 \times 10^8 = 1.875 \times 10^8 \text{ m/s}$$

Find refractive index of medium B

$$n_B = \frac{c}{v_B} = \frac{3 \times 10^8}{1.875 \times 10^8}$$
$$n_B = 1.6$$

8. Correct answer: B. Wavelength changes

Explanation:

- When light enters a new medium, its speed changes.
- Frequency remains constant (it depends on the source).
- Since $v = f\lambda$, a change in speed causes the wavelength to change.
- Energy does not become zero.

9. Correct answer: B $\boldsymbol{\frac{v_1}{v_2} = \frac{n_2}{n_1}}$

Explanation:

Refractive index is defined as:

$$n = \frac{c}{v}$$

where $c$is speed of light in vacuum and $v$is speed in the medium.

So,

$$v = \frac{c}{n}$$

For two media:

$$v_1 = \frac{c}{n_1}, v_2 = \frac{c}{n_2}$$

Taking ratio:

$$\frac{v_1}{v_2} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1}$$